Causal-Aware, Machine-Learning-Driven,

Factor-Informed Risk Forecasting

A Python Pipeline Integrating

NLP, Directed Factor Constraints, and Portfolio Analytics

Steven Archuleta

Andrew Nilsen

WorldQuant University

MScFE 690: Capstone Project

Submission Date: 01 June 2026

[GitHub Repository]

[MS Azure Dashboard]

Working Research Question:

How does imposing Manual Causal Constraints on
ML-driven Risk Forecasting Pipeline Models affect
Forecast Accuracy, Portfolio Performance, and Interpretability
compared to
Unconstrained Baseline Models?

Table of Contents

List of Figures

List of Tables

List of Abbreviations

The following abbreviations and short definitions cover the modeling, evaluation, factor, and macroeconomic terms used throughout this study.

Abbreviations	Expansion	Definition
ADF	Augmented Dickey-Fuller	Statistical test for stationarity
API	Application Programming Interface	Software interface for data access
Calmar Ratio	Calmar Ratio	(Annualized return) ÷ (max drawdown); a risk-adjusted return measure.
CART	Classification and Regression Trees	Decision tree algorithm used in XGBoost
CAUSAL_LSTM	Causal Constrained Long Short-Term Memory	An LSTM restricted to the 74 risk-stage features admitted by the manual DAG.
CAUSAL_MLP	Causal Constrained Multi-Layer Perceptron	An MLP restricted to the 74 risk-stage features admitted by the manual DAG.
CAUSAL_XGBOOST	Causal Constrained XGBoost	An XGBoost model restricted to the 74 risk-stage features admitted by the manual DAG.
CBOE	Chicago Board Options Exchange	Options exchange that publishes VIX
CI/CD	Continuous Integration/Continuous Deployment	Automated software development workflow
DAG	Directed Acyclic Graph	A flowchart of cause-and-effect in which information flows in one direction and never loops back.
DBC	Invesco DB Commodity Index Tracking Fund	Broad commodities ETF
DJIA	Dow Jones Industrial Average	Major U.S. stock market index
DM Test	Diebold–Mariano Test	Statistical test comparing the forecast accuracy of two models; tests whether the difference is significant.
DTB3	3-Month Treasury Bill Rate	The annualized yield on 3-month U.S. government debt; used as the risk-free rate proxy.
EEM	iShares MSCI Emerging Markets ETF	MSCI Emerging Markets ETF
EFA	iShares MSCI EAFE ETF	MSCI EAFE international developed markets ETF
ES	Expected Shortfall	Risk measure of average loss beyond VaR threshold
ETF	Exchange-Traded Fund	A basket of securities that trades on a stock exchange.
EWMA	Exponentially Weighted Moving Average	A volatility estimator that gives more weight to recent observations and less to older ones.
F1	F1 Score	Harmonic mean of precision and recall metrics
FinCARE	Financial Causal Analysis with Reasoning and Evidence	Hybrid causal discovery model combining algorithms with financial knowledge graph
FinReflectKG	Financial Reflective Knowledge Graph	Multi-step LLM reflection framework for financial knowledge graph construction
FRED	Federal Reserve Economic Data	Free public database of U.S. macroeconomic data published by the St. Louis Federal Reserve.
GARCH	Generalized Autoregressive Conditional Heteroskedasticity	A classical statistical model for forecasting time-varying volatility.
GES	Greedy Equivalence Search	Causal discovery algorithm

Abbreviations	Expansion	Definition
GLD	SPDR Gold Shares	Gold commodity ETF
H1	Hypothesis 1	Forecast Accuracy
H2	Hypothesis 2	Portfolio Risk Control
H3	Hypothesis 3	NLP Sentiment Contribution
HAC	Heteroskedasticity- and Autocorrelation-Consistent	Newey–West standard-error adjustment for overlapping time-series observations.
HML	High Minus Low (Value Factor)	A Fama–French factor capturing the value premium; the return difference between value and growth stocks.
HYG	iShares High Yield Corporate Bond ETF	High-yield corporate bond ETF
Invesco DB	Invesco DB Commodity Index Tracking Fund	broad commodities ETF
IWM	iShares Russell 2000 ETF	Russell 2000 small-cap equity ETF
KG	Knowledge Graph	Structured representation of relationships between entities
LLM	Large Language Model	AI language model used for text processing and reasoning
LQD	iShares Investment Grade Corporate Bond ETF	Investment-grade corporate bond ETF
LSTM	Long Short-Term Memory	A specialized neural network designed to remember patterns in sequential time-series data.
MACRO__	Macroeconomic Features	Feature prefix for macroeconomic indicators (DTB3, VIXCLS, T10Y2Y)
MAE	Mean Absolute Error	Average of absolute forecast errors; treats all errors equally regardless of size.
Max Drawdown	Maximum Drawdown	Largest peak-to-trough loss in portfolio value; measures worst-case loss experience.
Mkt–RF	Market Return minus Risk-Free Rate	Excess return of the broad market above the risk-free rate.
ML	Machine Learning	Algorithms that learn patterns from historical data to make predictions.
ML__	Machine Learning Features	Feature prefix for PCA latent factors
MLP	Multi-Layer Perceptron	A basic artificial neural network with input, hidden, and output layers.
MOM__	Momentum Features	Feature prefix for cumulative return momentum signals
MSCI	Morgan Stanley Capital International	Provider of global equity indices
MSCI EAFE	Morgan Stanley Capital International Europe, Australasia, and Far East	Index of international developed markets
NLP	Natural Language Processing	Using computers to read and interpret human language (news headlines).
NOTEARS	NOTEARS Algorithm	Continuous optimization approach for causal structure learning
OLS	Ordinary Least Squares	Standard linear regression; minimizes the sum of squared residuals.
PC	PC Algorithm	Causal discovery algorithm (Peter-Clark)

Abbreviations	Expansion	Definition
PCA	Principal Component Analysis	Reduces many correlated variables into fewer uncorrelated components capturing most variance.
QQQ	Invesco QQQ Trust	Nasdaq-100 technology-weighted equity ETF
REGIME__	Regime Features	Feature prefix for binary volatility regime indicators
ReLU	Rectified Linear Unit	Nonlinear activation function used in neural networks
RF	Risk-Free Rate	Return available with zero risk (3-month Treasury bill); benchmark for excess returns.
RMSE	Root Mean Squared Error	The square root of the average squared forecast error; penalizes large errors heavily.
S&P	Standard & Poor's	Credit rating agency and index provider
SEC 10-K	Securities and Exchange Commission Form 10-K	Annual regulatory filing by public companies
SENT__	Sentiment Features	Feature prefix for NLP-derived sentiment scores
Sharpe Ratio	Sharpe Ratio	(Annualized return minus risk-free rate) ÷ (annualized volatility).
SMB	Small Minus Big (Size Factor)	A Fama–French factor capturing the size premium; the return difference between small- and large-cap stocks.
SPDR	Standard & Poor's Depositary Receipts	ETF brand by State Street Global Advisors
SPY	SPDR S&P 500 ETF Trust	The most widely traded ETF, tracking the S&P 500 large-cap stock index.
SVAR	Structural Vector Autoregression	Econometric model for analyzing causal relationships
T10Y2Y	10-Year Minus 2-Year Treasury Spread	Yield-curve spread; negative values (inverted curve) often signal recession.
T-bill	Treasury Bill	Short-term U.S. government debt security
TLT	iShares 20+ Year Treasury Bond ETF	Long-term U.S. Treasury bond ETF
VAL__	Value Features	Feature prefix for Fama-French HML factor exposures
VaR	Value at Risk	Risk measure estimating potential losses
VAR-LiNGAM	Vector Autoregression with Linear Non-Gaussian Acyclic Model	Causal discovery method for time series
VIX / VIXCLS	CBOE Volatility Index	Chicago Board Options Exchange index measuring expected 30-day market volatility; the "fear gauge."
VOL__	Volatility Features	Feature prefix for EWMA and realized volatility measures
XGBoost	Extreme Gradient Boosting	A high-performance ML algorithm that builds many small decision trees sequentially.
XLE	Energy Select Sector SPDR Fund	Energy Select Sector ETF
XLF	Financial Select Sector SPDR Fund	Financial Select Sector ETF
XLK	Technology Select Sector SPDR Fund	Technology Select Sector ETF
XLV	Health Care Select Sector SPDR Fund	Health Care Select Sector ETF

Abstract

This project develops a causal-aware, machine-learning-driven pipeline for factor-informed financial risk forecasting. The central research question evaluates whether manual causal constraints, encoded as a nine-node seven-edge directed acyclic graph (DAG) restricting feature-to-model information flow on economic reasoning, improve out-of-sample forecast accuracy, portfolio performance, and interpretability relative to unconstrained baselines. The pipeline integrates daily prices and returns for 14 diversified ETFs, Fama–French three factors and the daily risk-free rate, FRED macroeconomic indicators (DTB3, VIXCLS, T10Y2Y), and NLP-derived news sentiment.

Three hypotheses guide evaluation. The first tests forecast accuracy via RMSE, MAE, and the Diebold–Mariano test with Newey–West HAC standard errors. The second tests portfolio risk control via Sharpe ratio, maximum drawdown, and volatility-stability targeting under a Ledoit–Wolf covariance estimator with a 10% annualized volatility target. The third evaluates the marginal contribution of NLP sentiment via a matched-window ablation. An interpretability diagnostic examines feature balance via normalized Shannon entropy and factor-exposure entropy. A three-stage ablation isolates each component's contribution; Stage 4 (full sentiment propagation across 2016–2025) is structurally precluded because the Alpha Vantage sentiment archive begins only in January 2020.

Our executed performance evaluation covers four unconstrained baselines (EWMA, XGBoost, MLP, LSTM) and three DAG-constrained causal models (CAUSAL_XGBOOST, CAUSAL_MLP, CAUSAL_LSTM) under a walk-forward expanding-window protocol with strict time-ordered splits and leakage governance. Hypothesis 1 is not confirmed: the DAG-constrained tree model incurs a moderate RMSE penalty relative to the unconstrained XGBoost baseline and EWMA, though Diebold–Mariano tests yield no statistically significant differences. Hypothesis 2 is partially confirmed: in the primary decision tree and statistical comparison, the DAG-constrained tree model achieves higher Sharpe ratio, shallower maximum drawdown, and higher Calmar ratio than both comparators. Hypothesis 3 is directionally supported by the matched-window sentiment ablation. The interpretability diagnostic confirms increased feature-importance and factor-exposure entropy under DAG constraints. Full reproducibility is ensured via a version-controlled Python package, fixed random seed, and continuous integration.

Keywords: causal inference, risk forecasting, directed acyclic graph, machine learning, factor-informed modeling, NLP sentiment, portfolio analytics, ablation study

Chapter 1: Introduction

1.1 Background and Motivation

Volatility modelling is an important part of portfolio risk management because it provides quantitative foundation for decisions about asset allocation, leverage sizing, drawdown hedging, derivative pricing, and regulatory capital requirements [1], [2]. Accurate volatility forecasting is an input for portfolio construction.

Two complementary paths dominate the volatility-modeling literature: a classical path that includes Exponentially Weighted Moving Average (EWMA), Generalized Autoregressive Conditional Heteroskedasticity (GARCH) family models, and Markowitz mean-variance optimization [3]; and a machine-learning path that includes gradient-boosted trees and neural-network architectures such as Multi-Layered Perceptron (MLP) and Long Short-Term Memory (LSTM) models. Depending on the application context, each path offers distinct advantages and limitations [2], [4], [5]. Both paths, however, inform our project, which evaluates a classical EWMA benchmark alongside ML baselines (XGBoost, MLP, LSTM) and DAG-constrained causal counterparts.

These two modeling paths translate directly into portfolio practice through diversification and risk budgeting, which remain foundational objectives in modern portfolio construction. Risk forecasts inform allocation decisions, leverage constraints, and drawdown-control mechanisms. The quality of risk forecasts depends on covariance and factor-exposure estimates that are, themselves, sensitive to the statistical properties of financial time series, including non-stationarity, volatility clustering, heavy tails, and time-varying correlation structure [6], [7]. Classical mean-variance optimization, introduced by Markowitz in 1952, always assumes a stable covariance matrix and it treats all sample covariance matrix entries as meaningful inputs [3]. Noise can dominate the in-sample covariance matrices under real market conditions, especially when the number of assets approaches the number of observations [6].

Machine-learning models help address the rigid structural assumptions of classical models through flexible nonlinear function approximations, capturing complex interactions among large feature sets. Relative to traditional GARCH-family estimators and realized-volatility estimators, gradient-boosted decision trees, random forests, and neural networks have demonstrated competitive risk-forecasting performance [8], [9], [10], [11]. Unconstrained machine-learning models face a well-documented hazard. With access to large feature sets, such models tend to exploit spurious in-sample correlations that lack economic justification. Bombari and Mondelli [12] document that unconstrained models can perform well in-sample yet degrade sharply during regime changes. The aforementioned degradation reflects a fundamental limitation shared by classical models and unconstrained ML approaches. Neither classical models nor ML models encode explicit assumptions about the direction of influence among economic variables, and both models' families treat features without distinguishing genuine economic mechanisms from coincidental statistical patterns [13], [14], [15], [16], [17].

The inability to incorporate genuine monetary impacts mechanisms makes such models particularly vulnerable to regime shifts and prone to producing unstable volatility forecasts [18], [19]. Such instability can directly affect portfolio risk by producing unstable allocation inputs, higher turnover/rebalancing pressure, and weaker factor-exposure attribution [6], [20], [21], [22]. Explainability is a core requirement for institutional investors. A forecast that lacks explanation through a coherent economic narrative is difficult to act upon and difficult to govern. The March 2020 COVID-19 market crash illustrates the problem as an abrupt shift, documented as a correlation-structure during that period caused unconstrained machine-learning models to exploit spurious nonlinear in-sample interactions that lack economic justification [23], [24], [25]. Timm [7] demonstrates that covariance-matrix noise degrades portfolio performance to the point where enhanced optimization methods are needed to recover stability, and Michel et al. [17] further show that, at scale, correlation-based factor models are fundamentally descriptive and cannot differentiate genuine causal factors from spurious correlations.

A growing body of literature in causal inference and factor investing argues that imposing domain-driven directional constraints on the modeling pipeline can mitigate this instability [16], [15], [13]. A causal approach encodes assumptions about how variables influence one another, rather than treating all features as exchangeable predictors. This specifies both the direction of each influence and the sequence in which those influences must operate. Directed acyclic graphs (DAGs) provide the formal structure for such constraints, establishing a topological ordering that restricts information flow and prevents shortcuts that violate economic reasoning, thereby improving forecast stability [17], [13], [14].

We discovered that the intersection of NLP-derived sentiment and factor-based risk modeling offers an additional avenue for incorporating modern information into risk estimation [27], [28]. Financial news and regulatory filings contain forward-looking language that can influence short-term momentum and risk perception [26], [17]. Integrating sentiment as a feature requires explicit decisions about whether text-derived signals enter the pipeline as direct risk predictors or operate upstream through momentum and/or returns, and about whether that placement remains stable across different market regimes [27], [15], [29], [30], [31].

Our project evaluates whether a small, manually specified directed acyclic graph[1] measurably improves out-of-sample risk forecast accuracy, portfolio-level risk control, and factor-exposure interpretability relative to an unconstrained machine-learning baseline, within a single reproducible pipeline.

1.2 Problem Statement

Unconstrained machine-learning risk models can exploit predictive relationships that do not reflect economically plausible information flow. Both the unconstrained gradient-boosted models and the linear regression models may assign spurious weights to correlations that are either coincidental, reverse-causal, or confounded by unobserved common drivers, especially in a high-dimensional feature space that combines price-based returns, macroeconomic indicators, factor exposures, and text-derived sentiment scores [18], [13]. Such relationships may appear predictive within the training window yet deteriorate out-of-sample when the data-generating regime changes [12], [18], [23], [24], [25].

The practical consequences are direct: unreliable volatility targets [2], [21], excessive portfolio turnover from unstable/noisy allocation inputs [6], [20], and reduced interpretability of factor-exposure attribution [13], [15], [16], [22]. A forecast that cannot be explained through a coherent economic narrative is difficult to act upon and difficult to govern by portfolio managers and risk officers [13], [15], [17].

A manual directed acyclic graph (DAG) provides a compact mechanism for addressing these problems [13], [14], [15], [17]. The DAG constrains the model's feature access by restricting which features are available to each modeling stage and enforcing a sequential estimation order, thereby reducing the opportunity for spurious pattern exploitation and aligning the pipeline with domain assumptions about causal direction [12], [13], [15], [18], [26], [27], [28]. Our primary question is whether these constraints produce measurable improvements in forecast accuracy, portfolio-level performance, and interpretability relative to a strong unconstrained baseline.

1.3 Research Question

The main research question guiding our study is:

> *How does imposing manual causal constraints on an ML-driven risk forecasting pipeline affect forecast accuracy, portfolio performance, and interpretability compared to unconstrained baseline models?*

This question is operationalized through a structured ablation design that progressively compares an unconstrained baseline, a matched-window NLP sentiment variant, and a DAG-constrained model variants. Our ablation design isolates the marginal contribution of each stage.

[1] Encodes economically motivated directional constraints over factor exposures, sentiment signals, and risk estimates

1.4 Research Hypotheses

Three formal hypotheses and one secondary interpretability diagnostic formulate our empirical evaluation. Each hypothesis maps to specific metrics and planned evidence objects (tables and figures) in later chapters.

H1 (Forecast Accuracy)

Claim: The DAG-constrained pipeline will reduce out-of-sample risk forecast error, measured by RMSE and MAE on 20-trading-day forward realized volatility, annualized as sqrt(252) × std(log returns over the next 20 days), relative to an unconstrained baseline trained under the same dates, target, split protocol, and hyperparameters, with the only experimental difference being feature access,[2] thereby isolating the effect of the DAG feature restriction. The primary H1 evidence comes from a 25-block expanding-window walk-forward evaluation conducted on the held-out test partition[3].

H2 (Portfolio Risk Control)

Claim: The DAG-constrained allocation will produce lower maximum drawdown and lower realized portfolio volatility during stress periods than the baseline allocation, while maintaining comparable return generation under the same volatility-targeting framework. The primary H2 evidence comes from a volatility-targeting portfolio backtest over the 499-trading-day active window (2023-12-29 through 2025-12-31), in which all seven model variants were evaluated under an identical rebalancing protocol, cost structure, and covariance estimator, thereby isolating the effect of causal versus unconstrained forecasting on portfolio-level risk outcomes

H3 (NLP Sentiment Contribution)

Claim: Adding NLP-derived sentiment features will improve near-term risk forecast accuracy relative to a no-text baseline. The primary H3 evidence comes from a matched-window evaluation on the post-2020 period (2020-01-03 through 2025-12-31, 1,496 rows), in which a sentiment-inclusive and a sentiment-excluded XGBoost model were trained and evaluated on identical dates and splits, thereby isolating the marginal contribution of the sentiment market feature.

1.5 Interpretability Diagnostic (Entropy-based)

Normalized Feature-importance entropy is measured as normalized Shannon entropy of XGBoost gain shares, which serves as a secondary diagnostic rather than a formal testable hypothesis. This diagnostic expects higher entropy under DAG constraints, indicating more balanced reliance across model features relative to the unconstrained model. Factor-exposure entropy computed from rolling Fama–French factor-loading regressions across all 24 rebalancing segments and provided portfolio-level evidence consistent with this diagnostic. Throughout our study, entropy is interpreted as a dispersion measure rather than an indicator of economic optimality. Higher entropy signals reduced concentration in a single dominant feature but does not guarantee improvement in risk-adjusted performance.

1.6 Scope and Delimitations

The scope of our project uses a daily-frequency pipeline operating over 2,798 trading days (from 2015-01-05 through 2026-02-19). The asset universe consists of 14 diversified ETFs: U.S. large-cap equity (SPY, QQQ), U.S. small-cap equity (IWM), international developed markets (EFA), emerging markets (EEM), sector equity (XLK, XLF, XLE, XLV), U.S. Treasuries (TLT), investment-grade credit (LQD), high-yield credit (HYG), gold (GLD), and broad commodities (DBC). Supplementary data sources include Fama–French daily factors (Mkt-RF, SMB, HML, RF), FRED macroeconomic series (DTB3, VIXCLS, T10Y2Y), and Alpha Vantage news sentiment article records that are aggregated into daily market-level sentiment scores.

[2] 151 unconstrained non-sentiment features for the baseline versus 74 DAG-allowed risk-stage features for the constrained model

[3] 2023-12-28 through 2025-12-31, 500 observations per ticker

The causal constraint layer is implemented as a small, manually specified DAG with the following directed edges:

Sentiment → Momentum → Returns

Sentiment Amplifies Momentum; Momentum Forecasts Returns

Modeling constraint: news tone is made available to the momentum estimation stage before momentum estimates enter the return forecast stage.

Value → Returns

Value (HML) Explains Returns

Modeling constraint: relative valuation signals contribute directly to expected return estimates.

Volatility → Risk → Allocation

Volatility Estimates Risk; Risk Constrains Allocation

Modeling constraint: observed volatility informs the forward-looking risk forecast, which then determines portfolio weights.

MACRO → Risk

Macro Conditions Inform Risk

Modeling constraint: macroeconomic indicators (VIXCLS, T10Y2Y, DTB3) are made available to the Risk estimation stage as exogenous conditioning variables.

REGIME → Risk

Regime Conditions Inform Risk

Modeling constraint: the binary VIX-regime indicator is made available to the Risk estimation stage as a volatility-state conditioning variable.

The DAG encodes economically motivated directional constraints rather than identified causal relationships. Each edge direction reflects a domain assumption about the order of information flow. Our project evaluates whether or not those directional constraints improve out-of-sample forecast accuracy and portfolio performance relative to unconstrained baseline models. The DAG is not presented as the true data-generating process. Several elements remain outside the scope of this study: full causal discovery at scale (PC, GES, or NOTEARS across hundreds of variables), full knowledge-graph construction from regulatory filings, multi-agent LLM causal reasoning systems, high-frequency trading execution, and complex multivariate GARCH models. These exclusions reflect the 10-week delivery timeline and the central methodological goal: to isolate the contribution of a transparent, auditable governance layer rather than obscure that contribution with a larger and less testable system.

1.7 Significance of our Study

Our study contributes an empirically evaluated template for integrating causal assumptions into a ML risk pipeline while preserving temporal integrity, governance transparency, and reproducibility. This project connects causal reasoning to practical portfolio analytics through a transparent ablation design that isolates the effect of sentiment features and DAG constraints on both forecast accuracy and allocation behavior. The pipeline demonstrates that a small, theory-driven DAG can serve as a lightweight governance layer. This constrains model inputs without requiring the computational overhead and sample-size demands of automated causal discovery. The factor-exposure entropy provides a quantitative measure of diversification quality that complements traditional portfolio performance metrics (Sharpe ratio, drawdown) and offers risk managers an additional diagnostic for monitoring allocation concentration.

Our project implements a version-controlled Python package, a modular notebook pipeline, unit tests for each modeling stage, and continuous integration via GitHub Actions. Results can be reproduced, validated, and extended. Our project charter and data dictionary provide documentation standards aligned with emerging expectations in financial ML governance.

Chapter 2: Theoretical Framework and Literature Review

Risk forecasting in modern portfolio construction sits at the intersection of Classical Portfolio Theory and Machine Learning, and the choice of how to combine them is a methodological decision rather than a stylistic one. Classical portfolio theory and the EWMA / HAR / GARCH family supply easily understood baselines, explicit assumptions, and interpretable failure modes. Machine-learning models supply nonlinear capacity, flexible feature use, and the ability to absorb large heterogeneous predictor sets. On its own, neither tradition supplies a disciplined account of which variables are allowed to influence which other variables, in what direction, and at what stage of estimation.

That gap is where our project attempts to bridge the literature. Our objective is to show why a causal-aware, machine-learning-driven, factor-informed risk forecasting pipeline is the correct object of interest and what each body of work contributes to that pipeline's design. Three distinctions guide our evaluations. Statistical association is not the same as economic interpretation; economic interpretation is not the same as causal identification; and a manually specified directed acyclic graph is not a claim about the true data-generating process. Throughout our project, the manual DAG is treated as a governance layer and an information-flow contract (rather than as a discovered causal map of financial markets).

2.1 Risk Forecasting in Equity Markets: Diversification, Covariance, and Classical Volatility Models

Risk forecasting matters in portfolio construction because allocation, leverage, hedging, and drawdown control all depend on credible expectations about future variability, rather than on realized variability alone. Academic literature treats volatility as a central state variable for derivative pricing, portfolio risk management, and volatility-managed allocation rules [2], [21]. For these reasons, classical risk models remain the baseline intellectual architecture against which machine-learning alternatives should be judged.

2.1.1 Modern Portfolio Theory and Diversification

Markowitz [3] formalized diversification as a portfolio-choice problem in which a decision maker selects portfolio weights to balance expected return against risk. Stated directly, attractive standalone assets do not automatically produce an efficient portfolio, because the interaction among assets matters as much as the behavior of each asset in isolation. Portfolio construction, therefore, begins with the weighted aggregation of expected returns.

$$E[R_{\mathrm{p}}] = w^{\mathrm{T}}\mu, \tag{1}$$

where $E[R_p]$ = expected portfolio return, w = vector of portfolio weights, μ = vector of expected asset returns, w^T = transpose of the weight vector. Although it is linear in weights, Equation (1) sets up a hard practical problem. The expected-return vector is unobservable and must be estimated from noisy data. This estimation challenge often shifts attention away from aggressive return forecasting and toward the more stable task of estimating risk and covariance structure [3], [21].

2.1.2 Portfolio Variance as a Covariance Problem

The more important quantity for risk forecasting is portfolio variance. A portfolio is not merely a collection of individual asset volatilities. A portfolio is a covariance system [3]. Diversification works when asset returns do not move together perfectly, and diversification can fail outright when correlations rise during stress episodes [3], [6].

$${\sigma_{\mathrm{p}}}^2 = w^{\mathrm{T}}\Sigma w, \tag{2}$$

where $\sigma_p{}^2$ = portfolio variance; w = vector of portfolio weights; Σ = covariance matrix of asset returns; $w^T\Sigma w$ = quadratic form measuring total portfolio risk. Total risk, in the quadratic form above, depends on both the diagonal elements of Σ, which capture individual asset variance, and the off-diagonal elements, which capture co-movement. The off-diagonal terms are decisive for a multi-asset ETF universe because sector ETFs, bond ETFs, broad-market

ETFs, and commodity ETFs respond differently to macroeconomic shocks. Ledoit and Wolf [6] show that covariance estimation becomes especially fragile when the number of assets is nontrivial relative to the available sample size.

2.1.3 Risk-Adjusted Performance and the Sharpe Ratio

Portfolio evaluation requires a risk-adjusted metric, and the Sharpe ratio remains the standard measure because it scales excess return by realized volatility; Sharpe ratio rewards return that are earned efficiently rather than returns earned through uncontrolled risk exposure [32].

$$Sharpe = \left(\frac{(\bar{r}_{\mathrm{p}} - \bar{r}_f)}{\sigma_{\mathrm{p}}}\right) \times \sqrt{252}, \tag{3}$$

where *Sharpe* = annualized Sharpe ratio; $\bar{r}_p$ = mean daily portfolio return; $\bar{r}_f$ = mean daily risk-free rate; σ_p = standard deviation of daily portfolio returns; $\sqrt{252}$ = annualization factor for approximately 252 trading days. The annualization in Equation (3) is an approximation that treats daily excess returns as additive over time and scales volatility by the square root of time. In practice, the Sharpe ratio is useful because the Sharpe ratio permits comparison among strategies that produce very different raw returns but operate at very different risk levels. The Sharpe ratio is best used when strategies produce very different raw returns at very different risk levels.

2.1.4 Classical Volatility Forecasting

Latent conditional volatility cannot be observed directly, so empirical finance typically works with realized-volatility proxies constructed from historical returns. Andersen and Bollerslev [33] show that realized return variation can serve as a practical proxy for latent volatility and can therefore support model evaluation. Our pipeline targets a forward realized-volatility series at a 20-trading-day horizon, which aligns naturally with a monthly-style rebalancing cadence.

$${\sigma^{20}}_{i,t} = \sqrt{252} \times std(r_{i,t+1}, r_{i,t+2}, \ldots, r_{i,t+20}), \tag{4}$$

where, ${\sigma^{20}}_{i,t}$ = 20-trading-day forward realized volatility for asset *i* at date *t*; $r_{i,t+k}$ = daily log return of asset *i* on day $t + k$; *std (·)* = sample standard deviation; $\sqrt{252}$ = annualization factor for approximately 252 trading days. EWMA, the classical **volatility model that our pipeline implements**, treats volatility as a weighted average of past squared shocks and gives larger weight to recent observations, which is why it survives as a benchmark in modern volatility studies [34].

$$\hat{\sigma}_t^2 = \lambda \hat{\sigma}_{t-1}^2 + (1 - \lambda) r_t^2, \hat{\sigma}_0^2 = r_0^2 \tag{5}$$

where $\hat{\sigma}_t^2$ = EWMA variance estimate at time *t*; λ = decay parameter; $\hat{\sigma}_{t-1}^2$= prior EWMA variance estimate; r_t^2 = squared return from time *t*; $1 - \lambda$ = weight on the most recent shock. Value-at-Risk and Expected Shortfall frameworks frequently use volatility forecasts as inputs, but our work focuses more narrowly on forward volatility prediction and portfolio risk control. The relevance of EWMA in our setting is, therefore, one of benchmark discipline. Any ML model must justify its added complexity by outperforming, or by delivering a compensating governance benefit relative to, this parsimonious classical alternative.

2.1.5 Why Classical Models Remain Necessary Benchmarks

Classical models remain necessary benchmarks because they provide transparent baselines, explicit assumptions, and interpretable failure modes. A machine-learning model that cannot outperform or meaningfully complement a disciplined classical baseline offers limited practical value. EWMA-style and covariance-based ideas therefore define the minimum standard against which our causal-aware machine-learning pipeline is judged.

2.2 Machine Learning in Financial Risk Forecasting

Machine learning enters financial risk forecasting because financial markets generate nonlinear, heterogeneous, and regime-dependent data. Price-based features, macroeconomic indicators, factor exposures, and sentiment signals can interact in ways that are difficult to specify a priori with low-dimensional parametric models. Supervised learning therefore recasts the risk-forecasting problem as the prediction of a future target from an input vector that is observable at the present date.

2.2.1 Supervised Learning as a Forecasting Problem

A forecasting rule in supervised learning is chosen from a hypothesis class by minimizing average loss on observed examples. The model is not estimating a single coefficient formula known in advance; instead, the model searches over a family of admissible functions and selects the one that best maps inputs to targets under a chosen loss criterion.

$$\hat{f} = \underset{f \in \mathcal{F}}{\operatorname{argmin}} \left(\frac{1}{N}\right) \sum_{i=1}^{N} L\big(y_{\mathrm{i}}, f(x_{\mathrm{i}})\big), \tag{6}$$

where $\hat{f}$ = fitted forecasting function; $\mathcal{F}$ = hypothesis space or model class; N = number of observations; $L(y_i, f(x_i))$ = loss function comparing realized target y_i and prediction $f(x_i)$; x_i = feature vector for observation i; y_i = realized target for observation i. The abstraction in Equation (6) is broad enough to cover linear regression, tree ensembles, feed-forward neural networks, and recurrent neural networks under one roof. The choice of loss function, hypothesis class, and validation protocol, therefore, determines how much flexibility, stability, and interpretability the final model can deliver.

2.2.2 Gradient Boosting and Tree-Based Nonlinear Approximation

Gradient boosting constructs a strong learner by adding weak learners sequentially, with each stage attempting to correct the residual structure left by earlier stages. Gradient-boosted trees are attractive because tree ensembles can model threshold effects, nonlinear interactions, and heterogeneous feature scales without extensive distributional assumptions. Empirical finance studies report strong volatility-forecasting performance for tree-based methods relative to classical alternatives, especially when predictor sets are rich and carefully governed [9], [11], [37].

$$\hat{f}^{(M)}(x) = \sum_{m=1}^{M} \gamma_{\mathrm{m}} h_{\mathrm{m}}(x)\,, \tag{7}$$

where $\hat{f}^{(M)}(x)$ = boosted prediction after M stages; γ_m = learning coefficient at stage m; $h_m(x)$ = weak learner or regression tree at stage m; x = input feature vector; M = total number of boosting stages. Equation (7) states that the final prediction is an additive ensemble rather than a single tree. That architecture gives tree boosting unusual flexibility on tabular datasets, but that same flexibility also creates a risk. A boosted model applied to a large and loosely structured feature set can exploit interactions that are statistically useful in-sample yet economically implausible out-of-sample.

2.2.3 MLP Models for Tabular Financial Data

A multi-layer perceptron model extends nonlinear approximation through stacked affine transformations and activation functions. An MLP transforms an input vector through hidden layers, with each hidden layer learning a new representation of the original features. For risk forecasting, an MLP can approximate complex nonlinear relationships in factor, macro, and sentiment spaces, but an MLP also depends heavily on feature scaling, regularization, and architecture choice. Small choices in any of those three areas can move out-of-sample results by a surprising margin.

$$h^{(l)} = \varphi\left(W^{(l)} h^{(l-1)} + b^{(l)}\right), \tag{8}$$

where $h^{(l)}$ = activations in layer l; $\varphi(\cdot)$ = nonlinear activation function; $W^{(l)}$ = weight matrix for layer l; $h^{(l-1)}$ = output from the previous layer; $b^{(l)}$ = bias vector for layer l. Equation (8) describes the representation-learning view of neural networks. Each layer remaps the incoming signal into a new feature space. However, this flexibility may also introduce instability when the signal-to-noise ratio is low and the data-generating regime changes.

2.2.4 LSTM Models for Temporal Dependence and Sequence Learning

Long Short-Term Memory networks address a central weakness of static tabular learners. Static tabular learners often treat each date as an isolated observation even though volatility is path dependent. An LSTM introduces a memory cell and gating structure that allow the network to retain, update, or discard information over time. That mechanism is relevant for volatility forecasting because volatility clustering, persistence, and shock decay are inherently sequential phenomena, and comparative finance studies therefore continue to test LSTM models alongside tree models and GARCH-family benchmarks [37].

$$f_t = \sigma\left(W_f[h_{t-1}, x_t] + b_f\right), \tag{9}$$

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i), \tag{10}$$

$$\tilde{c}_t = tanh(W_c[h_{t-1}, x_t] + b_c), \tag{11}$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t, \tag{12}$$

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o), \tag{13}$$

$$h_t = o_t \odot tanh(c_t), \tag{14}$$

where x_t = input vector at time t; h_t = hidden state; h_{t-1} = previous hidden state; f_t = forget gate; i_t = input gate; $\tilde{c}_t$ = candidate cell state; c_t = cell state at time t; c_{t-1} = cell state at time t-1; o_t = output gate; $\sigma(\cdot)$ = logistic sigmoid function; $tanh(\cdot)$ = hyperbolic tangent activation; $\odot$ = element-wise multiplication; W_f, W_i, W_c, W_o = gate weight matrices; b_f, b_i, b_c, b_o = gate bias terms. Equation (12) shows that an LSTM does not pass the full history forward unchanged. The forget gate filters old information, the input gate controls new information, and the output gate controls which memory state becomes the visible hidden state. This architecture can help with temporal dependence, but the architecture also introduces a heavier tuning burden, greater compute cost, and greater sensitivity to preprocessing choices.

2.2.5 Bias, Variance, Regime Sensitivity, and Spurious Pattern Extraction

Machine-learning models can reduce approximation bias because they are more flexible than low-dimensional parametric models. The trade-off is increased by estimation variance, which is amplified by non-stationarity, low signal-to-noise ratios, and the presence of many correlated candidate predictors in high-dimensional financial datasets. Spurious patterns can, therefore, appear predictive inside a particular sample window and then vanish once the regime changes [25], [12].

This instability creates a governance problem as much as a statistical problem; a model can appear accurate while relying on economically incoherent shortcuts or unstable interaction terms. Kaddour et al. [15] emphasize that causal machine learning is partly motivated by the need to discipline such failures, and empirical case studies document deterioration of unconstrained machine-learning systems during stress periods and structural breaks [24]. We introduce directional constraints that reduce unstable pattern extraction while retaining nonlinear forecasting capacity.

2.3 Causal Inference and Directed Acyclic Graphs

Causal graphical modeling offers a framework for reasoning about direction, mechanism, and admissible conditioning sets [13], [14]. In finance, the value of a DAG often lies less in proving intervention effects from observational market data and more in imposing disciplined restrictions on which relationships a model is allowed to learn [15]. Our manual DAG is used as a constraint system rather than as a claim that the true market causal network has been identified.

2.3.1 Foundations of Causal Graphical Models

Pearl [13] and Spirtes et al. [14] describe a directed acyclic graph as a graph whose nodes represent variables and whose arrows represent directional dependence assumptions, with acyclicity ruling out feedback loops that return to the starting node. A DAG encodes assumptions about conditional independence and about the admissible pathways through which information may travel.

$$P(X_1, X_2, \dots, X_\mathrm{n}) = \prod_{i=1}^{n} P(X_\mathrm{i} \mid Pa(X_\mathrm{i})), \tag{15}$$

where $P(X_1, X_2, \dots, X_n)$ = joint distribution over all variables; X_i = endogenous variable i in the graph; $Pa(X_i)$ = parent set of variable X_i; $\prod$ = product taken across all variables in the DAG. Equation (15) factorizes the joint distribution according to local parent sets. The factorization translates a large joint system into smaller conditional pieces. Once a parent set is specified for each variable, a researcher has also specified which variables are allowed to contribute direct information to each node.

2.3.2 Structural Causal Models and Parent Sets

A structural causal model goes one step beyond graphical notation by assigning each endogenous variable a functional dependence on its parents and on an exogenous disturbance term. Parent sets therefore do not merely organize visualization. They define the candidate information set that is permitted to influence each variable directly.

$$X_\mathrm{i} = f_\mathrm{i}(Pa(X_\mathrm{i}), U_\mathrm{i}), \tag{16}$$

where X_i = endogenous variable i in the graph; $f_i(\cdot)$ = structural assignment function for variable i ; $Pa(X_i)$ = parent variables of X_i; U_i = exogenous disturbance term or unobserved noise. Equation (16) is conceptually useful even when the functional form is not estimated literally [13]. The equation emphasizes that every modeled variable depends on a restricted local information set rather than on the entire observed dataset. That idea maps naturally into feature gating inside machine-learning pipelines [15].

2.3.3 *d*-Separation and Conditional Independence

d-Separation provides the graphical criterion for determining whether a path between two variables is open or blocked given a conditioning set. Chains and forks are blocked by conditioning on the middle node, whereas colliders behave differently and become informative only when the collider, or one of the collider's descendants, is conditioned upon [13], [14]. The lesson for empirical finance is sobering. More conditioning is not always better, and conditioning on the wrong variable can either destroy meaningful dependence or open a spurious path.

2.3.4 DAGs as Information-Flow Contracts in Finance

Real-world finance rarely permits clean intervention analysis, because financial markets are adaptive and most financial data are observational rather than experimental [13], [46]. A manual DAG is useful in such a setting, if the manual DAG is interpreted as an information-flow contract. Under that interpretation, a DAG specifies which types of features are economically reasonable inputs for each modeling stage and which direct connections should not be allowed [14]. The distinction between statistical association and economic interpretation matters. Observational financial data rarely supports formal causal identification, but a model still has to make structural assumptions if the goal is to reason beyond pure correlation. The model has to commit to some directional view of how variables relate. Encoding those commitments explicitly as a DAG makes the assumptions inspectable rather than buried in feature selection choices [15].

We adopt this interpretation in our project. Our DAG is small, theory driven, and stage based, and it does not claim to recover the true data-generating process. The DAG restricts feature access so that a risk model cannot learn every statistically available relationship. That choice turns the DAG into a governance regularizer rather than an intervention engine.

2.3.5 Manual DAGs versus Automated Causal Discovery

Automated causal-discovery methods, such as the PC algorithm and continuous-optimization approaches like NOTEARS, have expanded the methodological frontier for data-driven structure learning [14], [38]. Finance applications, such as D'Acunto et al. [16], show that evolving causal structures among equity risk factors can reveal regime-dependent sparsification and densification that ordinary correlation analysis misses. Richer systems, such as FinCARE [17] and FinReflectKG [39], illustrate how causal discovery can be combined with knowledge graphs, reasoning systems, and large-scale document processing.

These systems are valuable, but they also impose substantial sample-size, infrastructure, and governance demands. Many automated approaches stop at graph recovery or edge-quality evaluation and do not continue into portfolio-level empirical validation. Our manual DAG occupies a middle ground between unconstrained machine-learning prediction and infrastructure-heavy automated discovery. Our DAG is small enough to specify by hand, structured enough to bind feature access, and lightweight enough to test inside a 10-week research timeline.

2.4 Factor-Informed Modeling and the Fama–French Framework

Factor models remain central in financial economics because they decompose return variation into systematic components and idiosyncratic residual components [22], [40]. Factor theory plays a double role in a risk-forecasting pipeline. It supplies economically interpretable predictors and it provides a framework for auditing the exposures implied by portfolio construction.

2.4.1 Systematic versus Idiosyncratic Risk

Systematic risk refers to broad sources of variation that affect many assets simultaneously, whereas idiosyncratic risk refers to asset-specific variation that can often be diversified away in sufficiently broad portfolios [32], [22]. Factor models compress broad co-movement into a smaller set of drivers. That compression is useful because a factor representation is often more interpretable than a very large matrix of pairwise correlations, especially when the asset universe spans equities, fixed income, and commodities.

2.4.2 The Fama–French Framework

Fama and French [22] extended the Capital Asset Pricing Model by showing that equity returns can be described more effectively through a market factor, a size factor, and a value factor. The implication is that average returns differ not merely because securities have different standalone volatilities, but because securities load differently on systematic sources of return variation.

$$\mathrm{R}_{\mathrm{i},\mathrm{t}} - R_{f,t} = \alpha_i + \beta_{MKT,i}(Mkt - RF)_{\mathrm{t}} + \beta_{SMB,\mathrm{i}} SMB_{\mathrm{t}} + \beta_{HML,\mathrm{i}} HML_{\mathrm{t}} + \varepsilon_{i,t}, \tag{17}$$

where $R_{i,t}$ = return on asset i at time t; $R_{f,t}$ = risk-free rate at time t; α_i = asset-specific intercept or abnormal return; $\beta_{MKT,i}$ = market-factor loading for asset i; $(Mkt-RF)_t$ = market excess return at time t; $\beta_{SMB,i}$ = size-factor loading for asset i; SMB_t = size factor at time t; $\beta_{HML,i}$ = value-factor loading for asset i; HML_t = value factor at time t; $\varepsilon_{i,t}$ = idiosyncratic residual term. Newer factor-model research added profitability and investment factors, but the three-factor model remains analytically useful because it is compact and economically interpretable [40]. Factor exposures in a diversified ETF universe provide a bridge between raw return data and economically meaningful style characteristics.

2.4.3 Rolling Factor Exposure Estimation

Factor exposures are not static in practice. Asset composition changes over time, macro regimes shift, and market preference across sectors and styles changes as well. Rolling exposure estimation, therefore, translates static factor theory into dynamic measurement, and a rolling beta can be estimated directly from the ratio of return-factor covariance to factor variance.

$$\beta_{HML,i} = \frac{Cov(r_i, HML)}{Var(HML)}, \tag{18}$$

where $\beta_{HML,i}$ = value-factor loading for asset *i; Cov(*r_i*, HML)* = covariance between asset return and the HML factor; *Var(HML)* = variance of the HML factor; r_i = return series for asset *i.* The rolling beta captures time-varying value sensitivity while preserving a clear economic interpretation [22], [16]. The rolling beta in the feature pipeline is a univariate OLS slope estimated on raw log returns, and is therefore numerically distinct from the multivariate excess-return coefficient in Eq. (17). Our value node is operationalized through rolling HML exposures rather than through static balance-sheet ratios.

2.4.4 Covariance Estimation and Ledoit–Wolf Shrinkage

Portfolio construction ultimately requires a covariance matrix, not only a factor model. The textbook estimator is the sample covariance matrix, which becomes noisy and poorly conditioned when the estimation window is short relative to the number of assets. Ledoit and Wolf [6] address that problem by shrinking the sample covariance matrix toward a structured target, commonly a scaled identity matrix.

$$S = \left(\frac{1}{T - 1}\right) \sum_{t=1}^{T} (r_t - \bar{r})(r_t - \bar{r})^{\mathrm{T}}, \tag{19}$$

where S = sample covariance matrix; T = number of observations; r_t = asset returns at time t; $\bar{r}$ = sample mean return vector; $(r_t - \bar{r})^T$ = transpose of the centered return vector. Shrinkage deliberately introduces bias in exchange for materially lower estimation variance and better numerical stability. Timm [7] shows why that shrinkage logic matters operationally, noisy covariance estimates can dominate portfolio outcomes and can easily overwhelm improvements in the forecasting model itself. Covariance governance is, therefore, inseparable from forecast governance.

2.4.5 Entropy as a Concentration and Diversification Metric

Entropy from information theory provides a concise way to measure concentration or dispersion [41]. In our context, entropy does not measure economic welfare or portfolio optimality directly. It measures how evenly a total share is distributed across components such as feature-importance shares or factor-loading shares [45].

$$H = -\sum_{k=1}^{K} p_k ln(p_k), \tag{20}$$

where H = Shannon entropy; K = number of components, factors, or features; p_k = normalized share for component k; *ln(·)* = natural logarithm.

$$H_{norm} = \frac{H}{ln(K)}, \tag{21}$$

where H_{norm} = normalized Shannon entropy; H = raw Shannon entropy; K = number of nonzero components; *ln(K)* = maximum possible entropy under equal weighting. Normalization matters because raw entropy is mechanically affected by the number of available components. A constrained model with fewer features can exhibit lower raw entropy even if it uses the available features more evenly. The normalized form corrects that asymmetry and, therefore, supports more defensible cross-model comparison.

2.4.6 Why HML Is a DAG Node but Mkt-RF and SMB Are Not

Our DAG treats HML asymmetrically relative to Mkt-RF and SMB for a parsimony reason. HML receives explicit DAG status because value exposure varies meaningfully across our 14-ETF universe and offers a clear economic interpretation across broad-market, sector, and cross-asset instruments [22]. Mkt-RF, by contrast, is close to redundant for broad U.S. equity ETFs and would risk turning a small governance DAG into a near restatement of the equity market itself. SMB remains theoretically relevant but is retained as a systematic-risk control and decomposition variable rather than as an explicit DAG node [40].

That asymmetry is deliberate. A small DAG must choose which relationships are sufficiently interpretable and incremental to justify direct representation, and factor theory supplies both the predictor logic and the interpretability logic behind our value node, our covariance layer, and the entropy diagnostics that follow.

2.5 NLP Sentiment in Financial Forecasting

Financial text contains information about expectations, uncertainty, disagreement, and narrative tone that may not be fully reflected in price or factor data at the moment the text appears. Text matters in volatility forecasting because changing tone can precede repricing, risk re-assessment, and short-run shifts in market attention. The challenge is not only extracting a sentiment score. The challenge is placing the text signal in a pipeline with appropriate temporal governance and economic interpretation.

2.5.1 Why Financial Text Can Matter for Returns and Volatility

News headlines, filings, and market commentary can affect realized volatility through multiple channels. Financial text can alter expectations about cash flows, policy, credit conditions, litigation risk, or macro uncertainty. Loughran and McDonald [26] show that finance-specific terminology behaves differently from everyday usage, and more recent forecasting studies show that news analytics can contribute useful signal for volatility prediction, especially when text features are aligned carefully with trading calendars and forecast horizons [27].

2.5.2 Lexicon-Based Approaches and Finance-Specific Dictionaries

Early sentiment systems relied on dictionaries to classify words as positive, negative, uncertain, or litigious. The central issue was domain specificity. A word such as "liability" can be negative in everyday usage while remaining neutral in accounting context. Loughran and McDonald [26] address that problem by building finance-specific dictionaries that better capture the semantics of filings and market disclosures. Lexicon methods remain attractive because they are transparent and easy to audit, but they can miss context, negation, and cross-sentence meaning.

2.5.3 Transformer-Based Sentiment Models and Contextual Finance NLP

Recent finance NLP systems increasingly rely on contextual encoders and transformer-style architectures that can represent negation, semantic scope, and longer textual dependencies more effectively than dictionary counts alone. Causal-reasoning systems such as FinCARE [17] and knowledge-graph pipelines such as FinReflectKG [39] demonstrate that text can be integrated with richer reasoning and constraint frameworks rather than treated as an isolated standalone signal. We do not build an LLM-based reasoning system, but the literature still matters because it shows that textual information can be embedded in more structured financial pipelines.

2.5.4 Daily Sentiment Aggregation and One-Trading-Day Lag Governance

A practical ETF risk pipeline requires a daily market-level signal rather than thousands of raw article records. The natural aggregation step is a daily average sentiment score. Let $s_{j,t}$ denote the sentiment score of article j published on calendar day t, and let N_t denote the number of relevant articles on that day.

$$\tilde{S}_{\mathrm{t}} = \left(\frac{1}{N_{\mathrm{t}}}\right) \sum_{j=1}^{N_t} s_{\mathrm{j,\,t}}, \tag{22}$$

where $\tilde{S}_t$ = raw aggregate sentiment score for calendar day t; N_t = number of relevant news articles published on day t; $s_{j,t}$ = sentiment score for article j published on day t. To avoid look-ahead bias, the model cannot consume same-day sentiment that would not have been fully observable before the trading decision. The lag-governed feature used by the model must, therefore, shift the raw sentiment series by one trading day.

$$x_{\mathrm{t}}^{\mathrm{sent}} = \tilde{S}_{t-1}, \tag{23}$$

where x_t^{sent} = sentiment feature used by the model on trading day t; $\tilde{S}_{t-1}$ = aggregate sentiment score from the previous calendar or trading day. Equation (23) protects temporal integrity by ensuring that the feature used at date t is built only from information that was available before the trading session at date t. This governance step is not optional because same-day text can easily leak future information into the feature matrix [42].

2.5.5 Why Our Pipeline Uses Alpha Vantage News-Derived Market Sentiment

Our pipeline uses a lagged market-level sentiment series derived from Alpha Vantage news sentiment rather than contemporaneous firm-level text. That choice is aligned with an empirical setting; the target universe is a diversified multi-asset ETF panel, not a firm-specific event-study design, so a broad market-tone signal is more appropriate than firm-level text patterns. The Alpha Vantage archive also provides a practical bridge between raw article records and a reproducible daily feature, although the post-2020 archive boundary of Alpha Vantage scores constrains our full-history sentiment ablation.

Theoretical placement matters as much as feature extraction. A sentiment signal can be treated as a direct predictor, an upstream attention signal, or a regime-conditioning variable. Recent work on sentiment-aware volatility forecasting, including deep learning approaches with explicit sentiment integration [43], reinforces the importance of placement decisions inside the pipeline. Sentiment in our work is treated as an economically interpretable market signal whose use must be lag governed and structurally placed inside the pipeline rather than appended as a decorative add-on. Sentiment is evaluated as part of the economic information chain from narrative tone to market dynamics to risk rather than as an unconstrained text score.

2.6 Reproducibility and Governance in Financial ML

Reproducibility and governance are not auxiliary engineering details. They determine whether a backtest can be trusted at all. A forecasting pipeline can appear sophisticated while still failing on the most basic methodological grounds[4] [18].

2.6.1 Temporal Ordering and Leakage Avoidance

Forward-looking labels create a special leakage risk in volatility forecasting, because the target at date t is constructed from returns that occur after date t. Feature engineering, train-test splitting, and validation logic must, therefore, respect chronology at every stage. A leakage-safe training boundary can be expressed as follows.

$$t_{train,end} \leq t_{test,start} - forecast\_horizon, \tag{24}$$

where $t_{train,end}$ = final date included in the training set; $t_{test,start}$ = first date of the prediction or test block; *forecast_horizon* = forecast horizon in trading days. Equation (24) ensures that the training set ends at least *forecast_horizon* days before the prediction block begins. For a 20-trading-day forward volatility target, that gap prevents overlap between the label-construction window and the prediction window. A model cannot be allowed to learn from observations whose labels already contain the returns that will later be used to evaluate the forecast.

[4] If data leakage, temporal misalignment, hidden state, or uncontrolled experimentation enter the workflow

2.6.2 Walk-Forward Validation and Single-Use Holdout Logic

Walk-forward expanding-window validation is the counterpart to leakage-safe labeling in financial time series. Random cross-validation breaks temporal dependence and produces unrealistically optimistic estimates because future observations can influence the fitted model indirectly. An expanding window, by contrast, mimics the information set that would have been available to a portfolio manager at each forecast date [47]. The single-use holdout principle then reserves the final test period for one-time evaluation rather than repeated tuning [18]. Forecast comparison under overlapping horizons creates an additional inference issue because forecast errors become serially correlated. Diebold and Mariano [44] show why predictive-accuracy comparisons require tests that account for dependent loss differentials rather than naïve independent-sample tests. The broader governance is that valid empirical finance requires both a realistic validation scheme and an inference scheme that matches the dependence structure of the data.

2.6.3 Version Control, Artifact Contracts, and CI/CD Discipline

Reproducible financial ML also depends on disciplined software practice. Version-controlled code, pinned environments, modularized functions, saved intermediate artifacts, and automated test workflows reduce the role of hidden state and make the research process auditable. An artifact contract is especially valuable in research workflows because each stage can read a documented upstream artifact and write a documented downstream artifact, reducing ambiguity about how later tables and figures were generated [48], [49], [50]. CI/CD discipline extends that logic by automating linting, testing, environment checks, and build validation. Such discipline does more than improve software polish. It improves research credibility by narrowing the gap between a one-time result and a repeatable pipeline result [48], [50].

2.6.4 The DAG as a Lightweight Governance Layer

A governance-oriented interpretation of the DAG unifies the methodology. The DAG narrows the permitted search space, the walk-forward design preserves chronology, the artifact contract preserves traceability, and the validation framework preserves out-of-sample meaning. Relative to full knowledge-graph causal systems, a manual DAG can provide a lightweight governance layer that is interpretable, computationally feasible, and easy to audit [17], [39]. The theoretical payoff is controlled flexibility; it is not guaranteed accuracy. A model remains expressive, but it is prevented from learning every statistically available shortcut. We evaluate the causal-aware pipeline as a governed forecasting system rather than as a pure prediction artifact.

2.7 Research Gap and Contribution

The reviewed literature reveals a field that is mature in parts but still only partially integrated. Classical volatility models are well developed, machine-learning forecasting is increasingly competitive, factor models remain foundational, finance NLP is expanding, and causal graphical methods continue to evolve. The unresolved problem is integration under realistic governance [15].

2.7.1 What the Literature Does

The literature already provides strong building blocks. Classical volatility modeling offers benchmark discipline through EWMA, HAR, and GARCH families [10], [8], [35]. Machine-learning studies show that tree ensembles and neural networks can capture nonlinear structure in realized-volatility and related forecasting tasks [9], [19], [11]. Factor investing and covariance estimation research provide economically meaningful structure for interpreting returns and constructing portfolios [22], [6]. Finance NLP research shows that text can add incremental information when extraction and timing are handled carefully [26], [27]. Causal graphical modeling and causal-discovery literature provide a formal framework for directional assumptions and structure learning [13], [14], [38].

2.7.2 What Remains Missing

Many machine-learning risk papers are unconstrained and therefore weak on interpretability or governance. Several causal-structure papers stop at graph characterization and do not continue into portfolio-level empirical validation. A number of finance NLP studies evaluate text features in isolation rather than inside a full risk-allocation pipeline.

Reproducibility is also often underdeveloped; code, environments, and artifact lineage are frequently insufficiently documented for true end-to-end replication [18]. Recent work on entropy-augmented forecasting and portfolio construction at the industry-group level [45] points to the same integration gap.

2.7.3 Four Closely Related Studies

Four studies illustrate the gap. Each captures part of what our project tries to integrate, and each leaf something on the table.

FinCARE
Michel et al. [17] developed FinCARE, a hybrid model that combines PC, GES, and NOTEARS causal-discovery algorithms with a financial knowledge graph derived from SEC 10-K filings and LLM modules. Reported improvements across three causal-discovery baselines on synthetic data are striking. PC improves by +36%, GES by +100%, and NOTEARS by +366%. The FinCARE ablation design isolates the contributions of the knowledge graph, the LLM prompting, and the discovery algorithms. The framework is evaluated only against synthetic data. It stops at graph-recovery metrics such as F1, precision, and recall, with no portfolio-level validation.

FinReflectKG
Arun et al. [39] constructed FinReflectKG using SEC 10-K filings of S&P 100 index firms, applying a multi-step LLM reflection framework and obtaining a 64.8% compliance score that surpassed baselines using single-pass or multi-pass extraction approaches. As an open-source, large-scale financial knowledge graph developed from regulatory filings, FinReflectKG provides structured relationships that can serve as inputs for causal-discovery pipelines. Evaluation is limited, however, to KG quality metrics, with neither predictive ability nor portfolio performance assessed. A further limitation for downstream forecasting is that text-extracted edges may make causal claims without independently validating the underlying mechanism, and KG edges extracted from annual regulatory filings can incorporate lagging or backward-looking causal relationships that are not representative of current market dynamics.

Christensen et al.
The closest ML prediction benchmark in the volatility-forecasting space is Christensen et al. [9], which uses regularized methods, regression trees, and neural networks to predict realized volatility on DJIA stocks within the HAR family of models. The inclusion of structured predictors such as momentum, macroeconomic variables, and lags of realized volatility validates the feature classification used in our pipeline, while the empirical support for VIX and T-bill rates as macroeconomic predictors of volatility strengthens the rationale for the MACRO node in our DAG design. The crucial differences are that the Christensen et al. model assumes predictors to be exchangeable inputs, does not extend evaluation beyond MSE and QLIKE, is restricted to a single asset type (DJIA stocks), and lacks any interpretability metric.

D'Acunto et al.
D'Acunto et al. [16] conducted the first empirical study of dynamic causal relationships among 11 risk factors of U.S. equities based on SVAR coupled with VAR-LiNGAM in a sliding-window estimation approach, showing sparsification in causal relationships during normal regimes and densification during crises. Their result helps motivate why correlation-based models can degrade under regime shifts. The D'Acunto et al. causality graph serves as an endpoint, however, not as an input for a forecast or portfolio-management pipeline, and therefore no forecast-accuracy assessment is made. In addition, the VAR-LiNGAM identification assumption regarding non-Gaussian shocks may be violated in some regimes.

2.7.4 Contribution of Our Project

Our project addresses the middle ground between unconstrained ML forecasting and infrastructure-heavy automated discovery through three connected contributions. The first contribution is methodological. A small, manually specified DAG is used as a lightweight governance layer that restricts economically implausible information flow before model fitting begins, without requiring the sample-size demands and infrastructure overhead of automated discovery. The second contribution is empirical. Our pipeline integrates a diversified 14-ETF universe, factor-informed feature engineering, macro variables, and lag-governed market sentiment inside a structured ablation design that evaluates forecast accuracy, portfolio risk control, and interpretability together rather than as isolated objectives.

The third contribution is governance oriented. The pipeline is framed as a reproducible end-to-end workflow with version-controlled code, modular artifacts, and machine-learning operations discipline, thereby offering a transparent template for defensible financial ML research. Taken together, the literature supports the conclusion that the theoretical ingredients of our work are individually established but only partially integrated in prior work. The framework also addresses three audiences. Quantitative portfolio managers requiring interpretable models under governance constraints; Risk officers conducting causality analysis for regulatory and stress-testing purposes; and Machine-learning finance practitioners seeking an operational benchmark codebase. Our work, therefore, positions itself as a middle-ground contribution between unconstrained ML forecasting and infrastructure-heavy automated causal-discovery systems; we are intentionally smaller and more auditable, and empirically tied to forecast accuracy, portfolio performance, and interpretability metrics. Chapter 2 has established the conceptual logic of our project. Classical portfolio theory explains why covariance structure and volatility estimation matter; machine-learning literature explains why nonlinear models can improve forecasting while remaining vulnerable to spurious relationships, causal graphical modeling explains how directional constraints can regularize that vulnerability, factor theory explains how systematic exposures can be interpreted; NLP literature explains how text can enter a market-risk pipeline, and governance literature explains why reproducibility is a methodological requirement rather than a packaging preference. The next chapter operationalizes those commitments within one unified pipeline, defining the data, features, target, DAG contract, model families, and evaluation design.

Chapter 3: Data and Methodology

3.1 Research Design Overview

Our project employs a quasi-experimental ablation design that targets four stages but executes three. We test DAG features gating on the full window non-sentiment pipeline through stage 1 and stage 3 comparison, while Stage 2 tests sentiment on the matched post-2020 window only.

Stage 1 is the 151-feature non-sentiment XGBoost baseline model trained on the full 2016–2025 window.

Stage 2 is a matched-window post-2020 sentiment comparison between sentiment-augmented XGBoost and non-sentiment-augmented XGBoost.[5] The matched-window design isolates sentiment's marginal contribution within the post-2020 period but cannot be directly compared to Stage 1 on equal footing.[6]

Stage 3 is the DAG-constrained model family: CAUSAL_XGBOOST, CAUSAL_MLP, and CAUSAL_LSTM.

Stage 4 is structurally unavailable. Stage 4 would have provided a full incremental ablation propagating sentiment through the complete 2016–2025 feature matrix and evaluating all four stages against a common training window. The third-party Alpha Vantage sentiment data archive begins in January 2020. No pre-2020 sentiment series exists within our current data pipeline, and no pipeline revision can resolve this constraint without sourcing an alternative sentiment archive with pre-2020 coverage. This Stage 4 constraint is a vendor data-coverage boundary common across the global research community that uses Alpha Vantage data for academic research, not a design failure. The ablation ladder enables each stage to be evaluated against the immediately preceding stage, such that the performance deltas can be attributed to a single added component, rather than to the joint effect of multiple simultaneous changes. All ablation stages share the same risk forecast target, 20-trading-day forward realized volatility, annualized, as well as the same expanding-window validation protocol, the same rebalancing frequency, and the same transaction cost convention. These conventions ensure that observed performance differences reflect model quality rather than backtest design artifacts.

[5] Alpha Vantage sentiment archive begins in 2020, so a full-history 2016-2025 Stage 2 is structurally unavailable.
[6] The Stage 1 versus Stage 3 comparison, which tests the core causal constraint question, is unaffected by this limitation and remains the primary empirical contribution of our study.

3.2 Risk Forecast Target Definition

The risk forecast target for every ETF in the asset universe is the 20-trading-day forward realized volatility, computed as the annualized standard deviation of daily log returns over the subsequent 20 trading days. Formally, for ETF i at date t, the target $\sigma^{20}{}_{i,t}$ is defined as:

$$\sigma^{20}{}_{i,t} = \sqrt{252} \times std(r_{i,\,t+1}, r_{i,\,t+2}, \ldots, r_{i,\,t+20}) \tag{25}$$

where $r_{i,\,t}$ denotes the daily log return of ETF i on date t, computed as $ln(P_{i,t} / P_{i,t-1})$ from adjusted close prices. The standard deviation uses Bessel's correction *(ddof = 1)* and the $\sqrt{252}$ annualization factor converts the daily standard deviation to an annualized volatility metric consistent with industry convention for approximately 252 trading days per calendar year.

The target definition was frozen on 2026-02-24 and used consistently across all hypothesis tests, metric tables, and evaluation narratives in our project. Our feature engineering implements the target construction as a two-step shift operation: first shifting log returns by -1 to align index t with the return at $t + 1$, then computing a rolling 20-day standard deviation, and finally shifting the result by -19 to re-align the volatility estimate to the decision date t. The resulting target dataset contains 2,798 rows × 14 columns, with 20 trailing missing observations per ticker corresponding to dates where the forward window extends beyond the available sample.

3.3 Data Sources and Collection

3.3.1 ETF Price Data

The asset universe consists of 14 diversified exchange-traded funds (ETFs) selected to span multiple asset classes and economic sectors: SPY (S&P 500 large-cap equity), QQQ (Nasdaq-100 technology-weighted equity), IWM (Russell 2000 small-cap equity), EFA (MSCI EAFE international developed markets), EEM (MSCI Emerging Markets), XLK (Technology Select Sector), XLF (Financial Select Sector), XLE (Energy Select Sector), XLV (Health Care Select Sector), TLT (iShares 20+ Year Treasury Bond), LQD (iShares Investment Grade Corporate Bond), HYG (iShares High Yield Corporate Bond), GLD (SPDR Gold Shares), and DBC (Invesco DB Commodity Index Tracking Fund). Daily adjusted close prices for all 14 ETFs were downloaded via the *yfinance* Python library. The visualization of the ETF price data is shown in Figure 1.

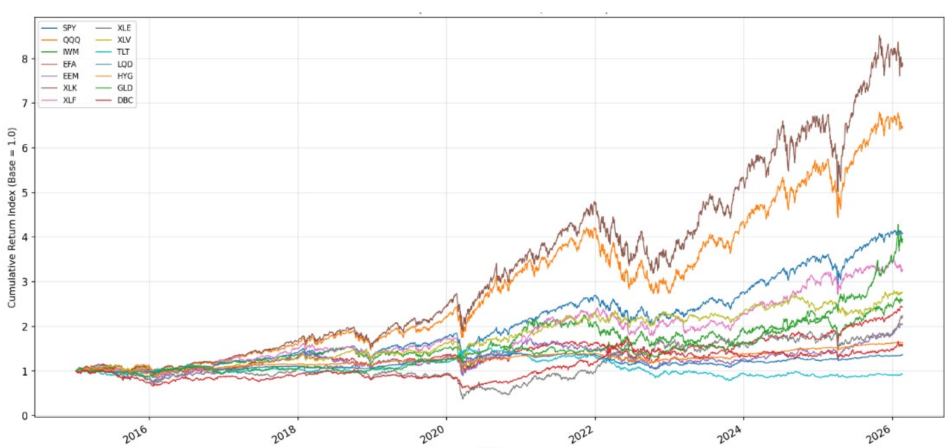

Figure 1. Cumulative returns for all 14 ETFs over the 2015–2026 sample period.

3.3.2 Macroeconomic Indicators

Three macroeconomic indicator series were sourced from the Federal Reserve Economic Data (FRED) database: DTB3 (3-Month Treasury Bill secondary market rate, daily), VIXCLS (CBOE Volatility Index closing value, daily), and T10Y2Y (10-Year minus 2-Year Treasury constant maturity spread, daily). The visualization of the CBOE Volatility Index data is shown in Figure 2.

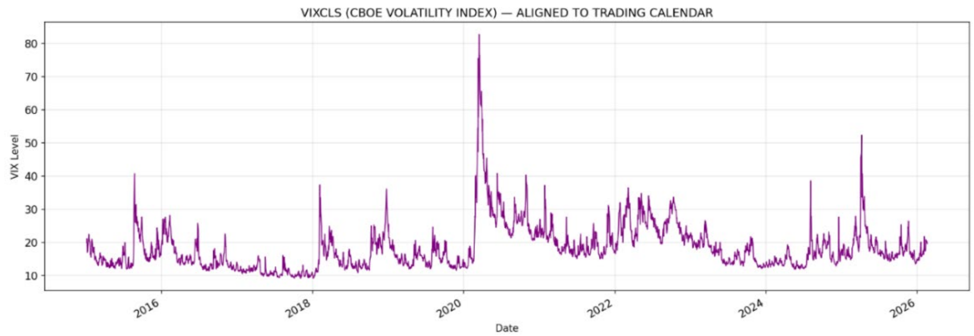

Figure 2. VIXCLS (CBOE Volatility Index) aligned to the ETF trading calendar over the full sample period.

3.3.3 Fama–French Factors

Daily Fama–French three-factor data (Mkt-RF, SMB, HML) plus the risk-free rate (RF) were downloaded from the Kenneth French Data Library.

3.3.4 NLP Sentiment Features

NLP-derived sentiment features were sourced from the Alpha Vantage API. The pipeline fetches article-level sentiment records that are aligned with financial market topics. Adaptive recursive window splitting with a maximum depth guard (*max_depth=8*) [7] prevented infinite recursion on high-volume date ranges during Q4 2025 and Q1 2026 earnings periods. The details of the data are shown in Table 1.

Table 1. Dataset Description, Treatment Summary, Feature Engineering Roles.

Data	Frequency	Sample Coverage	Aligned Rows	Missing Data & Treatment	Role in Pipeline
ETF Price	Daily	2015-01-02 – 2026-02-19	2,799 (prices); 2,798 (returns)[8]	None. Log returns computed as $r_{i,t} = \ln(P_{i,t}/P_{i,t-1})$	Cross-asset volatility modelling; ablation across equity, fixed-income, commodity, and sector allocations
Macro-Economic Indicators	Daily	2015-01-02 – 2026-02-19	2,798 (ETF calendar)	21 rows (~0.75%); left join on ETF calendar then forward-fill; no further imputation for residual gaps	VIXCLS → binary regime detection node (REGIME), provides binary high & low volatility indicator; T10Y2Y → term-structure dynamics relevant to risk regime transitions; DTB3 → daily risk-free rate for Sharpe ratio and portfolio analytics
Fama–French Factors	Daily	2015-01-02 – 2025 (latest update)	2,765 / 2,798 (98.8%)	33 rows in Jan–Feb 2026 due to known publication lag; no imputation applied; left join on ETF calendar, not forward-filled; converted to decimal form	Factor exposure controls
NLP Sentiment Features	Daily	2020-01-01 – 2026-03-18	458,662 unique articles, covering 1,540 of 1,541 post-2020 trading days (99.9%)	Pre-2020 period (1,257 rows) absents by design; 2020-01-02 absent by construction; trading days with no headlines treated as missing, not forward-filled; computed as mean overall sentiment score of all articles published on calendar day t – 1; left joined on ETF calendar;	Sentiment signal for H3; one-trading-day lag enforced to prevent look-ahead bias; test partition (2023-12-28 – 2025-12-31) achieves 100% coverage

[7] A depth of 8 was chosen because each level halves the date window, so eight halvings split a multi-year request down to roughly a single day — fine-grained enough to fully capture even the densest earnings-week news volume, while still stopping the recursion before it can loop forever on a rate-limited or article-saturated window.
[8] One row fewer due to the differencing operation.

3.3.5 Exploratory Analysis and Stylized Facts

The effective window spans 2016-01-04 to 2025-12-31 (N = 2,496 rows). It is the period in which all features and the forward target are simultaneously non missing values. All training, evaluation, and EDA operations are restricted to this mask.

Table 2. Stylized facts summary for representative ETF returns and volatility behavior.

Stylized Fact	Evidence Summary	Modeling Implication
Effective Modeling Window Exists Where Features and Target Are Simultaneously Non-Nan	The sample starts on January 4, 2016, ends on December 31, 2025, and includes 2,496 observations.	All EDA, training, and evaluation operations are restricted to the effective modeling mask.
Log Returns Exhibit Stationarity Under *Adf* for Most Tickers	The log-return series is stationary at the 5% significance level for 100% of the observations.	Modeling based on return-derived features is empirically more justified than using price levels.
Forward Realized Volatility Target Exhibits Series-Dependent Stationarity Behavior	The 20-day forward annualized volatility series is stationary at the 5% significance level for 100% of the observations.	Robust metrics and regime-aware reporting are recommended when modeling volatility target dynamics.
Returns Distributions Exhibit Heavy Tails and Outlier Frequency Above Normal Baseline	The median excess kurtosis is 4.955205, and the mean three-sigma tail rate is 0.011619.	RMSE should be complemented by MAE, and optional *VaR/ES* reporting is motivated by tail risk.
Squared Returns Exhibit Serial Correlation Consistent with Volatility Clustering	At lag 20, the Ljung–Box test on squared returns rejects the null hypothesis for 100% of the sample.	The EWMA baseline and volatility-based feature families are empirically justified.
Cross-Asset Correlations Indicate Non-Trivial Co-Movement Structure	Cross-asset correlations indicate a non-trivial co-movement structure, with a mean absolute off-diagonal correlation of 0.465391.	Discussions of diversification and regimes should reference correlation clusters.

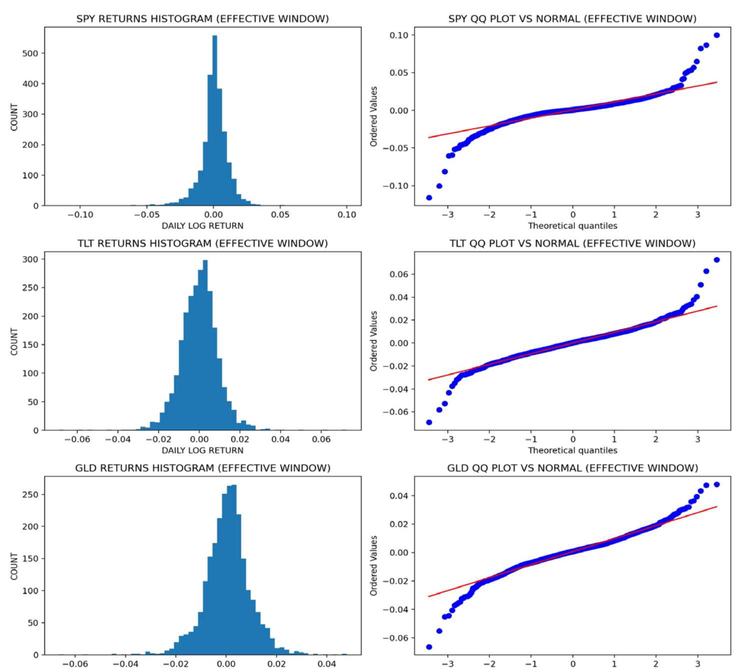

Figure 3. Distribution diagnostics for representative ETFs (SPY, TLT, GLD) — return histograms (left) and QQ plots versus normal (right). Heavy tails and positive excess kurtosis are evident in all three assets.

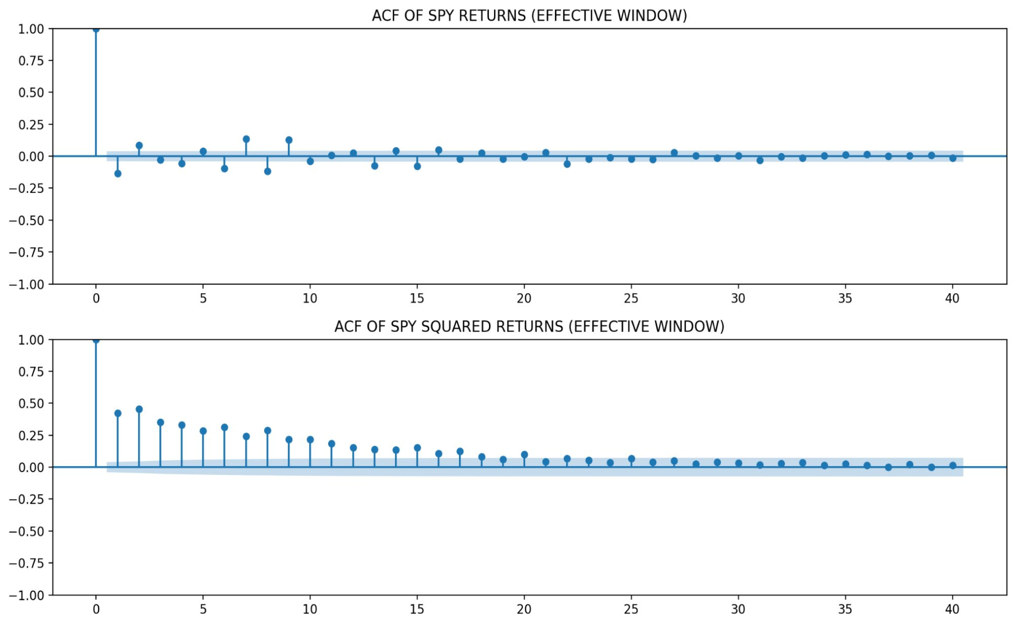

Figure 4. ACF of SPY returns (top) and squared returns (bottom). Returns show negligible serial correlation; squared returns show strong persistence consistent with volatility clustering.

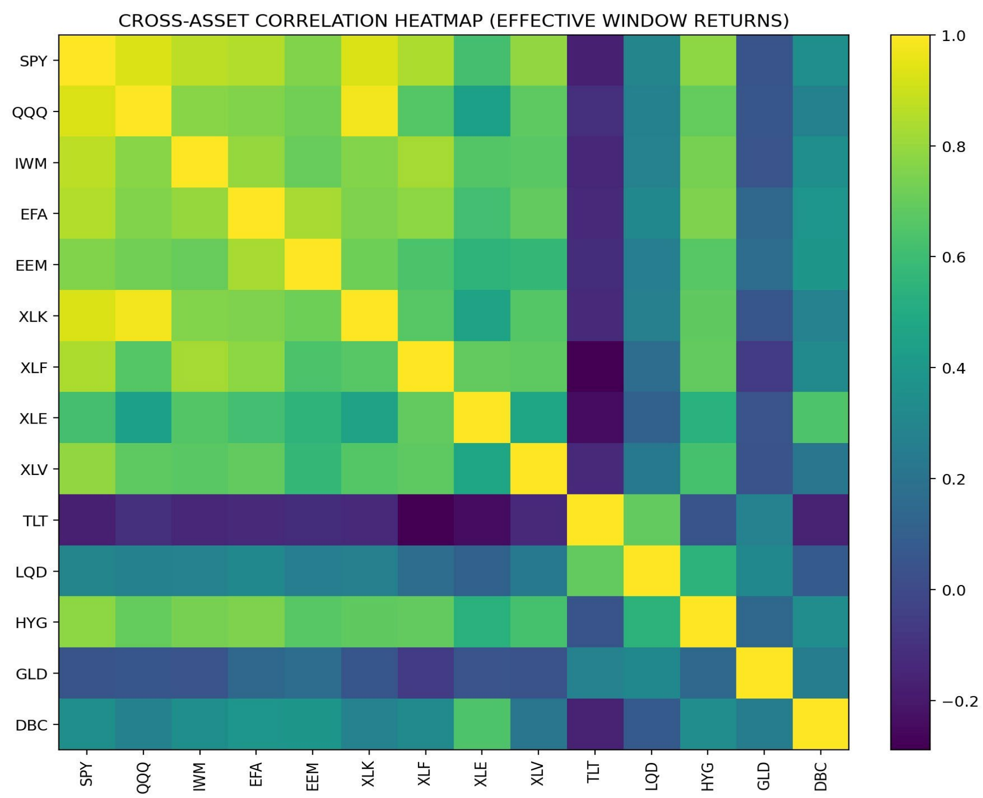

Figure 5. Cross-asset correlation heatmap for the 14-ETF effective modeling window. Equity ETFs form a high-correlation cluster (r ≈ 0.6–0.9); TLT and GLD show negative-to-zero equity correlation.

3.4 Feature Engineering

Engineered features are organized by seven DAG-node prefixes, with a total of 152 features in the full panel and 151 features in the main non-sentiment working matrix:

Table 3. Feature Prefix Families and DAG Risk-Stage Gate (Bold = Allowed in DAG).

Prefix	Count	Feature Family	Risk-Stage Status	Rationale
VOL__	70	EWMA volatility is computed with spans of 21 and 63, while realized volatility is measured over 10, 21, and 63 days using a rolling standard deviation (ddof = 1) and annualized by √252.	**Allowed**	Direct parent of Risk node
MACRO__	3	DTB3, VIXCLS, and T10Y2Y are calendar-aligned and forward-filled.	**Allowed**	Exogenous risk-environment conditioners
REGIME__	1	A binary VIX regime indicator is constructed by comparing the VIXCLS level to its expanding median with a minimum of 252 observations; dates where VIX exceeds this threshold are labeled 1.0 (high volatility), and all others are labeled 0.0 (low volatility).	**Allowed**	High/low volatility-state conditioner
MOM__	56	Cumulative log returns are computed over 5 days to capture short-term trend dynamics and over 10, 21, and 63 days to represent medium- to quarterly momentum signals.	Blocked	Upstream of Returns, not direct Risk
VAL__	14	14 Rolling 63-day HML beta features and 4 market-level Fama–French pass-through features are estimated via OLS, with β defined as the covariance between returns and the factor divided by the factor's variance.	Blocked	Value explains Returns, not direct Risk
ML__	3	Rolling PCA latent factors are estimated using a 252-day window, with a fixed random state seed of 692 to ensure reproducibility.	Blocked	No direct DAG parent role at Risk node
SENT__	1	Daily NLP sentiment score (Jan 2020–)	Blocked / H3 only	Coverage-limited and upstream of Momentum

The prediction target is 20-trading-day forward realized volatility is calculated with Equation (25). The target is physically separated from feature construction to protect against look-ahead leakage.

Table 4. Feature counts by DAG node prefix.

DAG Prefix	Feature Count
MACRO_	3
ML_	3
MOM_	56
REGIME_	1
SENT_	1
VAL_	18
VOL_	70
Total	**152**

3.5 Manual DAG Architecture

The central methodological contribution of our project is a small, manually specified Directed Acyclic Graph (DAG) that encodes economically motivated directional constraints on the modeling pipeline. The DAG restricts which features are available to each modeling stage and enforces a sequential estimation order aligned with domain assumptions about the plausible direction of information flow among factors, sentiment signals, and risk estimates. The DAG does not claim to represent the true data-generating process. A DAG functions as an information-flow contract and governance regularizer that reduces the search space for spurious relationships [13]. Our seven-edge DAG is applied as follows:

Table 5. DAG Edge Specifications and Economic Interpretations (9 nodes, 7 directed edges).

DAG Edge	Direction	Economic Meaning
SENTIMENT → MOMENTUM	Upstream input	News sentiment influences short-term momentum dynamics before momentum influences returns
MOMENTUM → RETURNS	Upstream input	Trend continuation signals influence returns
VALUE → RETURNS	Upstream input	Fama-French value factor betas influence returns
VOLATILITY → RISK	Primary input	Observed volatility influences forward-looking risk forecast
MACRO → RISK	Exogenous conditioning	Yield spread, rate level, and VIX influence the risk environment
REGIME → RISK	Exogenous conditioning	Binary VIX-regime indicator captures high/low volatility state influencing the risk environment
RISK → ALLOCATION	Downstream output	Forward risk predictions drive inverse-volatility portfolio weights

Visualized as a structural diagram, the same nine nodes and seven edges separate into a Risk-Allocation cascade and a Return-forecast cascade. Our manual DAG with nine nodes and seven directed arrows organized into three causal chains as shown in Figure 6.

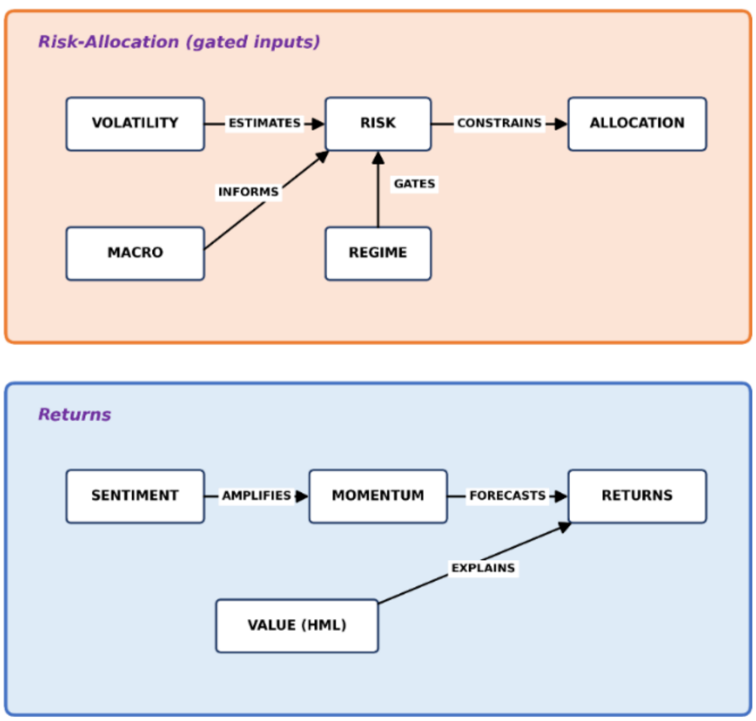

Figure 6. Manual DAG architecture showing nine nodes and seven directed arrows.

Sentiment → Momentum → Returns. The first chain encodes the assumption that text-derived sentiment influences momentum dynamics before momentum estimates enter the return forecast stage. News tone and headline polarity (SENT_ node) are made available to the momentum estimation stage (MOM_ node) as parent features. Momentum features, such as cumulative log returns over 5-day, 10-day, 21-day, and 63-day windows, then feed into the return forecast stage (RETURNS node) alongside the value features. A directional constraint prevents sentiment from bypassing momentum to predict returns directly. Enforcing this economically motivated assumption that sentiment operates on returns primarily through momentum channels rather than as an independent return predictor aligns with the financial causal-graph evidence reported in [17], [39].

Value → Returns. The second chain encodes the assumption that relative valuation signals contribute directly to expected return estimates. The Value node (VAL_) is operationalized through the Fama–French HML (High Minus Low) factor: rolling 63-day OLS betas of each ETF against the HML factor capture time-varying value exposure [22]. The raw Fama–French factor values (Mkt-RF, SMB, HML, RF) also pass through as VAL_ prefixed columns for systematic-risk decomposition. HML earns explicit DAG representation because the value premium varies meaningfully across ETFs with different value/growth tilts, making HML a defensible predictor without circularity concerns.

Volatility → Risk → Allocation. The third chain encodes the assumption that observed historical volatility informs the forward-looking risk forecast, which in turn determines portfolio weights. The Volatility node (VOL_) contains realized-volatility and EWMA-volatility features computed from historical returns. The Risk node receives volatility features as inputs and produces the 20-day forward realized volatility forecast. The Allocation node receives the risk forecast and determines portfolio weights through volatility targeting or risk-parity logic. The directional constraint prevents allocation decisions from influencing risk estimates or volatility calculations. This temporal ordering that reflects the physical sequence of portfolio management decisions.

The MACRO_ node contains the three FRED macroeconomic series (VIXCLS, T10Y2Y, DTB3), which function as exogenous conditioning variables available to all downstream modeling stages.

The REGIME_ node contains the binary high-volatility indicator (vix_low, vix_high), which is derived from the expanding median of the VIX series and serves as a regime-conditioning variable for stress-aware evaluation.

The asymmetric treatment of Fama–French factors in the manual DAG reflects a deliberate parsimony decision. HML earns explicit node status (Value → Returns) because the value premium captures a distinct cross-sectional return pattern that varies meaningfully across the ETF universe. The Market Excess Return (Mkt-RF) and Small Minus Big (SMB) factors do not appear as explicit causal nodes. Including Mkt-RF as a predictor of broad-market ETF returns introduces near-perfect multicollinearity: SPY returns correlate approximately 99% with the Mkt-RF series, which renders a Market → Returns edge as redundant for the primary ETF universe and weakens the "small DAG" interpretability claim. The exclusion of Mkt-RF and SMB from the manual DAG does not imply exclusion from the unconstrained pipeline. Both factors remain available as systematic-risk controls and future factor-decomposition candidates. The Asymmetric Fama-French Factor treatment in the manual DAG is visualized by Figure 7.

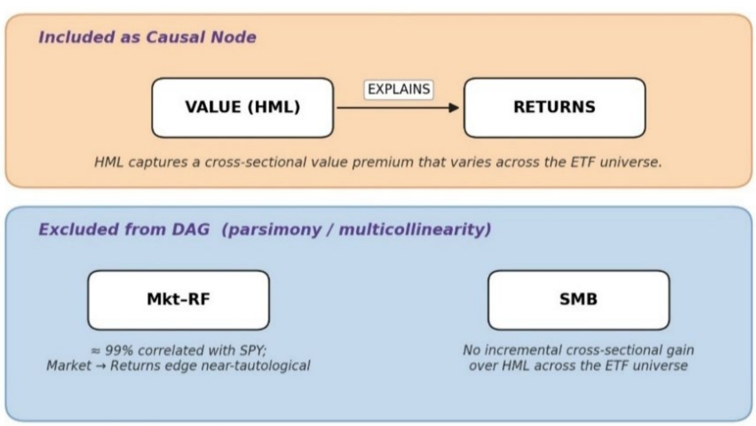

Figure 7. Asymmetric Fama-French Factor Treatment in the Manual DAG.

The DAG constraint is implemented operationally through prefix-based feature gating. Each modeling stage receives only the feature columns whose DAG-node prefixes appear in the allowed-parent set for that stage. The Momentum stage receives SENT_, MACRO_, and REGIME_ prefixes. The Returns forecast stage receives MOM_, VAL_, MACRO_, REGIME_, and ML_ prefixes, with SENT_ excluded because sentiment cannot bypass momentum to predict returns directly, and with VOL_ excluded because volatility is routed to the Risk stage rather than the Returns stage. The Risk forecast stage receives VOL_, MACRO_, and REGIME_ prefixes, with MOM_, VAL_, SENT_, and ML_ features excluded to prevent return-forecast information from leaking into the risk track. The Allocation stage receives only the risk forecast output; all raw features are excluded.

3.6 Modeling Framework

3.6.1 Baseline Models (EWMA, XGBoost, MLP, LSTM)

The unconstrained baseline model family consists of four variants: EWMA, XGBOOST, MLP, and LSTM. The machine-learning baselines receive the 151-feature matrix[9] without any causal restrictions on feature access. EWMA uses only the span-63 exponentially weighted moving standard deviation of daily log returns and does not consume the feature matrix. An MLP baseline and an LSTM baseline[10] are also trained and evaluated on the same 151-feature matrix using identical walk-forward protocols, extending the unconstrained model family to four variants: EWMA, XGBOOST, MLP, and LSTM. All four baselines establish the performance reference against which DAG-constrained models are evaluated.

EWMA Baseline

The EWMA forecast uses the span-63 exponentially weighted moving standard deviation of daily log returns, annualized by the √252 factor, as a naive volatility forecast at each date [34]. The EWMA estimator requires no training and serves as the minimum-viable benchmark: any ML model that cannot beat the EWMA forecast on out-of-sample RMSE and MAE provides no marginal value over simple exponential smoothing [8], [10]. The EWMA forecast is computed per ticker using the same annualization convention as the target definition.

The EWMA estimator gives more weight to recent observations and less weight to distant observations. In the *RiskMetric*s convention used in this pipeline, EWMA estimates volatility recursively by updating a variance estimate from the previous variance estimate and the most recent squared return, as shown in Equation (26):

$$\hat{\sigma}_t^2 = \lambda\hat{\sigma}_{t-1}^2 + (1-\lambda)r_t^2, \hat{\sigma}_0^2 = r_0^2, \tag{26}$$

where $\hat{\sigma}_t^2$ is the EWMA variance estimate at time t, λ is the decay parameter, r_t is the asset returns at time t, and the recursion is initialized at $\hat{\sigma}_0^2 = r_0^2$. The decay parameter $\lambda = 0.94$ places 94% weight on the previous variance estimate and 6% weight on the most recent squared return, and the model requires no training because the recursion has a closed form [10].

Inside the broader feature pipeline, four decay parameters $\lambda \in \{0.86, 0.91, 0.94, 0.97\}$ are applied across multiple rolling windows to construct 70 VOL_ features (Section 3.4); the pure EWMA baseline model used for forecast comparison applies $\lambda = 0.94$ per the RiskMetrics convention.

XGBoost Baseline

The unconstrained XGBoost model is trained on a 151-feature baseline matrix formed by excluding sentiment data. Hyperparameter selection follows a walk-forward validation protocol using the validation partition only, where the test set is never touched during tuning [47], [18]. Six candidate parameter sets were evaluated across four representative tickers (SPY, TLT, GLD, HYG) as shown in Table 6.

[9] 152 columns minus the excluded sentiment data
[10] Sequence length is 20 and hidden size is 32.

Table 6. Hyperparameters candidates of the XGBoost model.

Hyperparameters	Values	Explanation
n_estimators	*(400, 500, 600)*	Number of boosting trees
max_depth	*(3,4)*	Maximum depth of each decision tree
learning_rate	*(0.03, 0.05)*	Shrinks the contribution of each new tree
subsample	*(0.70, 0.80)*	Fraction of training observations randomly sampled for constructing each tree
colsample_bytree	*(0.80, 0.90)*	Fraction of predictor variables randomly sampled for each tree during training
reg_lambda	*1.0*	L2 regularization parameter for leaf weights
tree_method	*hist*	Algorithm used for tree construction and optimization during training

SPY, TLT, GLD, and HYG were selected as one representative per major asset class (equities, rates, commodities, credit) to capture cross-sector volatility diversity at a fraction of the compute cost. Each candidate was scored by the average RMSE across up to 8 walk-forward validation blocks per ticker, with a label-maturity lag of 20 trading days enforced to prevent any overlap between training labels and prediction blocks. The XGBoost model is a gradient-boosted tree ensemble in which *num_trees* additive regression trees (CART) are built sequentially, with each tree correcting the residual errors of all previous trees [51]. The goal is to minimize prediction error while penalizing model complexity to prevent overfitting. The prediction model of XGBoost is defined by Equation (27) as the sum of *num_trees* independent tree functions evaluated on the input feature vector.

$$\hat{y}_i = \phi(z_i) = \sum_{k=1}^{num\_trees} f_k(z_i), f_k \in \mathcal{H}, \tag{27}$$

where $\mathcal{H}$ is the space of regression trees, f_k is an independent tree structure, z_i is the feature vector for training instance i, *num_trees* is the total number of trees, and $\hat{y}_i$ is the predicted value. During training, XGBoost minimizes prediction error while keeping each tree simple through the regularized learning objective $\mathcal{L}(\phi)$ in Equation (28).

$$\mathcal{L}(\phi) = \sum_i \ell(\hat{y}_i, y_i) + \sum_k \Omega(f^k)\,, \Omega(f) = \gamma J + (½)\eta \left|\left|\kappa\right|\right|^2, \tag{28}$$

where ℓ is the loss function[11], J is the number of leaves in a tree, κ is the leaf weight vector, γ and η are regularization hyperparameters, y_i is the actual value, $\Omega(f)$ is the regularization term, and $\Omega(f^k)$ is the complexity penalty. Direct minimization of Equation (28) is intractable; Chen and Guestrin apply a second-order Taylor expansion to reduce the problem to a quadratic that can be optimized one tree at a time. The XGBoost model evaluates every candidate split point when building a tree and accepts a split if the prediction error improves by at least γ; otherwise, the node is left as a leaf [51].

Table 7. XGBoost hyperparameter tuning results

Rank	*n_estimators*	*max_depth*	*learning_rate*	*subsample*	*colsample_bytree*	Avg Val RMSE	Wall Sec
1	400	4	0.05	0.80	0.80	0.052448	59.7
2	600	4	0.03	0.80	0.80	0.052604	85.2
3	400	3	0.05	0.80	0.80	0.052786	36.0
4	500	3	0.05	0.70	0.90	0.053902	55.7
5	600	3	0.03	0.80	0.80	0.054076	53.2
6	500	4	0.05	0.70	0.90	0.055165	93.4

The best-performing parameter set achieved a validation RMSE of 0.052448. The RMSE spread across all six candidates was narrow (0.052448 to 0.055165, a range of 5.2%), indicating that the model is not highly sensitive to the tested hyperparameter variations. This is a positive sign for baseline robustness.

[11] mean-squared error is utilized in this pipeline

Walk-forward test evaluation applied the best parameter set to all 14 tickers. For each ticker, an expanding-window protocol trained XGBoost on all available data up to a label-safe boundary[12], predicted the next 20-day block, advanced the window, and repeated across all 25 test blocks. The expanding-window design ensures that the training set grows monotonically, mimicking the information set available to a portfolio manager at each rebalancing date.

MLP Baseline

The Multilayer Perceptron (MLP) is a feedforward neural network consisting of an input layer, one or more hidden layers, and an output layer, in which each neuron in one layer connects to every neuron in the next layer. This dense connection structure allows MLP to learn nonlinear relationships between inputs and outputs that linear models cannot capture. The training goal is to find weight matrices and bias vectors for all connections so that model output approximates the desired output by minimizing the difference between predicted and actual values [52], [53]. The MLP architecture used in this pipeline is described through three-layer types.

A. Input Layer. The input layer (h^0) of MLP receives the raw feature vector without transformation, defined by Equation (29):

$$h^0 = x, \tag{29}$$

where x is the input feature vector containing the 151 unconstrained features for BASELINE_MLP or the 74 DAG-allowed features for CAUSAL_MLP, and h^0 is the input-layer activation vector.

B. Hidden Layer. Each hidden layer applies a learned linear transformation followed by a nonlinear activation function. The hidden-layer transformation is defined by Equation (30):

$$h_l = \tau(W_l h_{l-1} + b_l), l = 1,2, \dots, L-1, \tag{30}$$

where h_l is the hidden layer l, W_l is the weight matrix for layer l, h_{l-1} is the hidden layer l-1; b_l is the bias vector for layer l, and $\tau(\cdot)$ is the activation function[13]. Each hidden layer applies weights and the activation function to learn nonlinear patterns in the data.

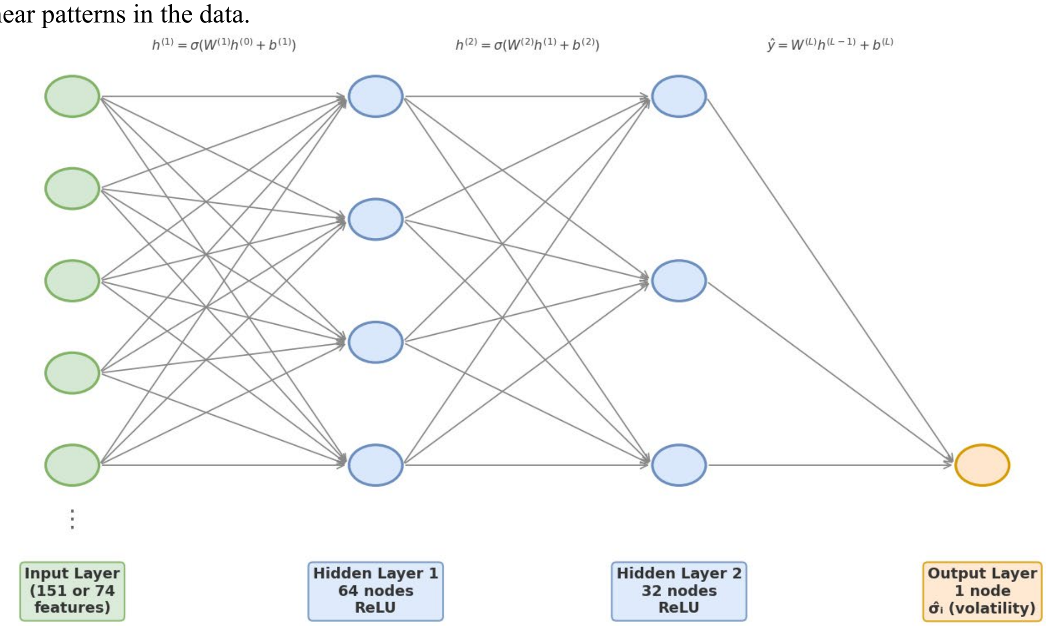

Figure 8. MLP Architecture.

[12] enforces the 20-day maturity lag
[13] ReLU in this pipeline

The architecture of MLP utilized in this project is as follows:

- BASELINE_MLP: the neural network starts with 151 input variables, processes them through hidden layers containing 128 and 64 neurons, and finally produces 1 prediction output. with ReLU activations on hidden layers;
- CAUSAL_MLP: the causal neural network uses 74 input variables, compresses the information through 64 and 32 neurons, and outputs 1 final prediction.
- Output node returns $\hat{\sigma_i}$, annualized 20-trading-day forward realized volatility.
- Both networks are trained with the Adam optimizer and early stopping with up to *max_iter* = 500 epochs (BASELINE_MLP) or *max_iter* = 400 epochs (CAUSAL_MLP).

C. Output Layer. The output layer maps the final hidden representation to a single forecasted volatility value through a linear transformation, with no activation function applied for regression tasks, as defined by Equation (31):

$$\hat{y} = W_L h_{l-1} + b_L, \tag{31}$$

where $\hat{y}$ is the predicted output, W_L is the output-layer weight matrix, b_L is the output-layer bias vector, and h_{l-1} is the hidden layer *l-1*.

LSTM Baseline

A recurrent neural network (RNN) learns temporal sequences by propagating error backwards through time, but suffers from gradient vanishing and exploding problems during backpropagation. Long Short-Term Memory (LSTM) addresses this by introducing a memory cell with a fixed self-connection that enforces constant error flow, together with three gates that control what information enters, leaves, and is forgotten by the memory cell [54]. The LSTM cell architecture used in this pipeline consists of six components.

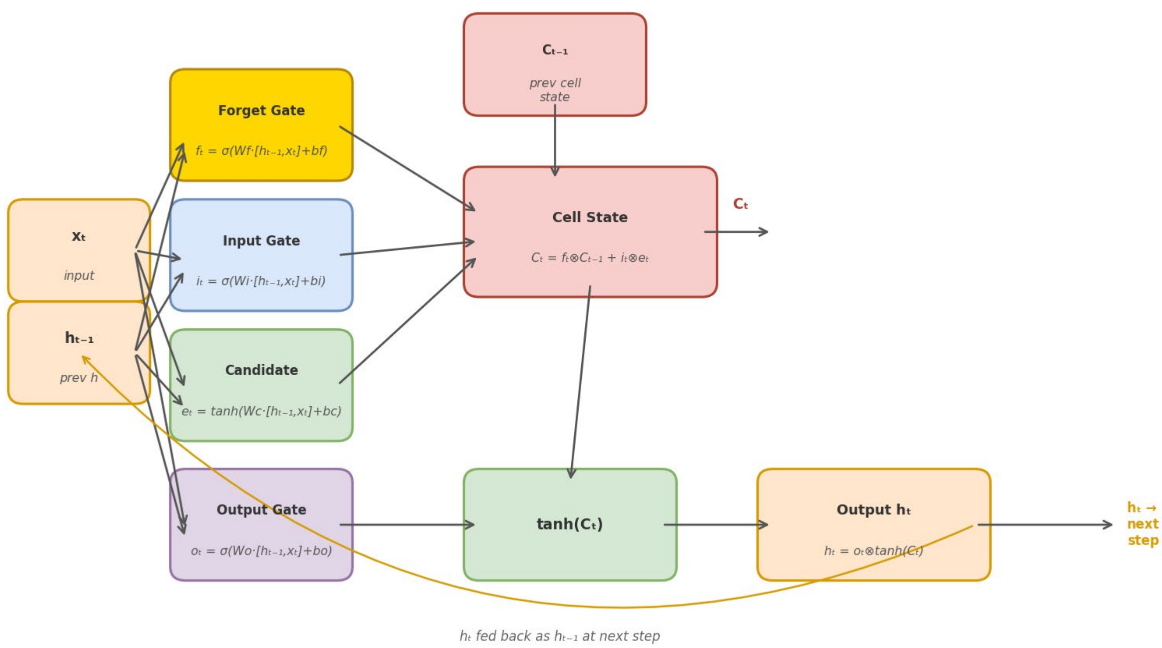

Figure 9. LSTM Architecture.

A. Forget Gate. The forget gate determines what fraction of the prior cell state to retain. Equation (32):

$$f_t = \sigma\big(W_f \cdot [h_{t-1}, x_t] + b_f\big), \tag{32}$$

where f_t is the forget gate output at time $t \in (0,1)$, $\sigma(\cdot)$ is the sigmoid activation function, W_f is the forget-gate weight matrix, $[h_{t-1}, x_t]$ is the concatenation of the previous hidden state and the current input vector, and b_f is the forget-gate bias vector.

B. Input Gate. The input gate determines what fraction of the candidate cell-state update to write into memory. Equation (33):

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i), \tag{33}$$

where i_t is the input-gate output at time t, σ is the sigmoid activation function, W_i is the input-gate weight matrix, and b_i is the input-gate bias vector.

C. Candidate Gate. The candidate gate generates the proposed new content for the cell state. Equation (34):

$$\tilde{c}_t = tanh(W_c \cdot [h_{t-1}, x_t] + b_c), \tag{34}$$

where $\tilde{c}_t$ is the candidate-gate output at time t, *tanh* is the hyperbolic tangent activation function, W_c is the candidate-gate weight matrix, and b_c is the candidate-gate bias vector.

D. Cell State. The cell state updates by combining the retained portion of the prior cell state with the gated candidate update. Equation (35):

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t, \tag{35}$$

where c_t is the updated cell state at time *t*, representing the long-term memory of the network after deciding what to forget and what to add, c_{t-1} is the cell state at time *t-1*, and $\odot$ denotes elementwise multiplication.

E. Output Gate. The output gate determines what portion of the new cell state becomes the visible hidden-state output. Equation (36):

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o), \tag{36}$$

where o_t is the output-gate value at time *t*, σ is the sigmoid activation function, W_o is the output-gate weight matrix, and b_o is the output-gate bias vector.

F. Output Layer. The hidden-state output combines the output gate with a *tanh*-modulated cell state. Equation (37):

$$h_t = o_t \odot tanh(c_t), \tag{37}$$

LSTM receives both the current data vector and the previous hidden state, combines the two, and passes the combined vector into all four gates simultaneously. The forget gate determines what portion of old memory to discard; the input gate and candidate gate together determine what new information to write into memory; the cell state is updated accordingly; and the output gate controls what portion of the updated cell state becomes the hidden-state output. The hidden state h_t serves both as the output of the current time step and the memory input to the subsequent step [54]. BASELINE_LSTM and CAUSAL_LSTM use a sequence length of 20 trading days, *hidden_size* = 32, and a dense output layer producing one annualized volatility forecast per ticker per date.

The Long Short-Term Memory (LSTM) baseline introduces explicit temporal structure that the XGBoost and MLP models lack. XGBoost and MLP treat each trading day as an independent observation with input shape (N, 151). The LSTM requires the same tabular data to be restructured into a three-dimensional sequence tensor of shape (N, 20, 151): each prediction target is informed by a rolling window of 20 consecutive daily feature vectors rather than a single-row snapshot, allowing the network to learn how volatility signals evolve over time.

The transformation is implemented utilizing *StandardScaler*, which fits on training rows only inside each walk-forward block before the sequence tensor is constructed, preserving leakage discipline. The LSTM architecture is a single-layer with hyperparameters: *hidden_size*=32, *head_dropout*=0.10, *learning_rate*=0.001, *batch_size*=64, *max_epochs*=35, and *early stopping with patience*=5 on an internal 10% validation tail. Walk-forward test evaluation ran across all 14 tickers and 25 test blocks using the same label-safe boundary protocol as XGBoost and MLP.

3.6.2 Causal-Constrained Model

The causal-constrained model applies the DAG architecture described to restrict feature access at each modeling stage. The constrained XGBoost model uses the same hyperparameter configuration as the unconstrained baseline to ensure that any observed performance differences are attributable to the causal feature restriction rather than to hyperparameter tuning differences. The only modification is the feature set: at each modeling stage, the constrained model receives only the feature columns whose DAG-node prefixes appear in the allowed-parent set for that stage.

The current implementation operationalizes the DAG at the Risk stage through prefix-gated feature access; the broader DAG remains the conceptual information-flow contract for the full pipeline. The risk forecast stage receives only VOL_ features[14], MACRO_ features[15], and REGIME_ features[16]. Momentum, value, sentiment, and ML-latent features are excluded from the risk forecast stage because the DAG does not permit information flow from those nodes into the Risk node. The constrained feature set totals 74 features versus 151 in the unconstrained baseline, a 51% reduction that serves as an implicit regularization mechanism: fewer features reduce the opportunity for the model to exploit spurious correlations that do not reflect economically plausible information flow [16]. Causal-constrained models are implemented using the identical hyperparameter configuration, expanding-window walk-forward protocol, and label-safe boundary as unconstrained baselines.

CAUSAL_MLP

The CAUSAL_MLP model applies the same architecture used in the un-constrained MLP baseline to the DAG-constrained 74-feature matrix[17]. Three hyperparameter candidates were evaluated, with hidden layer sizes specified for each configuration, where alpha was set to 0.0001, the learning rate initialization to 0.001, and the maximum number of iterations to 400; the candidates and their average validation RMSEs were:

Table 8. Hidden layer sizes and average RMSE of hyperparameter candidates.

Hyperparameter Candidates	Hidden layer sizes	Average RMSE
Candidate 1	(64,)	0.228305
Candidate 2	(64,32)	0.151962
Candidate 3	(128,64)	0.153805

The best *hidden layer sizes* is (64,32), achieved the lowest average validation RMSE of 0.151962.

CAUSAL_LSTM

The causal constrained LSTM model applies the same architecture used in the unconstrained LSTM baseline to the DAG-constrained 74-feature matrix. The input sequence tensor shape is (N, 20, 74): each prediction target is informed by a rolling window of 20 consecutive daily causal feature vectors. The architecture is identical to BASELINE_LSTM[18], ensuring that any performance difference between causal constrained LSTM and unconstrained LSTM is attributable solely to the DAG feature gate and not to architectural differences. Walk-forward test evaluation ran across all 14 tickers and 25 test blocks using the same label-safe boundary protocol as all other model families.

[14] 70 columns: realized volatility and EWMA volatility across three rolling windows and two exponential spans for each of the 14 ETFs
[15] 3 columns: *VIXCLS*, *T10Y2Y*, *DTB3*
[16] 1 column: *vix_high*
[17] VOL, MACRO, REGIME prefixes only
[18] *hidden_size*=32, *seq_len*=20, *head_dropout*=0.10, *learning_rate*=0.001, *batch_size*=64, *max_epochs*=35, *patience*=5

3.6.3 Ablation Ladder

The ablation ladder is a structured experimental design that progressively adds pipeline components to isolate the marginal contribution of each stage. The four ablation stages are:

Stage 1: Unconstrained Baseline. The full 151-feature XGBoost model with no causal restrictions (Section 3.6.1). The Stage 1 model establishes the performance reference against which all subsequent stages are compared.

Stage 2: NLP Sentiment added[19]. The sentiment feature is excluded from the baseline models training to preserve the full 2016-2025 training window. A matched-window H3 evaluation was executed on post-2020 dates only, training both Sentiment augmented XGBoost and non-sentiment augmented XGBoost on identical splits.

Stage 3: DAG Constraints added. The Stage 1 unconstrained XGBoost baseline already includes the full non-sentiment feature set[20]. Stage 3 is therefore the DAG-constrained model, which restricts the Risk stage to the 74 VOL, MACRO, and REGIME features. The Stage 3 versus Stage 1 comparison is the primary test of Hypothesis H1. A true four-stage ablation that incrementally adds ML factors at Stage 3 would require rebuilding Stage 1 without those features; this falls outside the scope of this study.

Stage 4: Full Sentiment Ablation. A genuine Stage 4 would require a sentiment series covering the full 2016–2025 window so that sentiment and non-sentiment models could be compared on an identical training history. That sentiment series does not exist because the Alpha Vantage archive begins in January 2020. It is structurally unavailable. The current executed comparison, Stage 1 versus Stage 3, represents the complete achievable ablation within this pipeline. Future work requiring a Stage 4 comparison would need to source an alternative financial news archive with continuous coverage from at least 2015.

Stages 1 and 3 share a common portfolio backtest protocol[21]. Stage 2 is forecast-only on the matched post-2020 window and is not portfolio-comparable; Stage 2 portfolio metrics are reported as NA in the ablation table. The ablation table reports RMSE, MAE, maximum drawdown, Sharpe ratio, and factor-exposure entropy for each stage.

3.7 Portfolio Construction and Backtesting

A refactored volatility-targeting portfolio construction protocol is implemented, where we convert the 20-day forward realized-volatility forecasts produced by seven model variants[22] into parallel monthly-style backtests on a common out-of-sample calendar. The backtesting window covers the 500 common test-partition trading dates from 2023-12-28 through 2025-12-31; because trades execute on the next trading day, the active return series begins on 2023-12-29 and contains 499 realized portfolio-return observations per model. The common calendar is partitioned into 24 rebalancing events at 21-trading-day intervals. The primary H2 narrative continues to emphasize the tree/statistical comparison[23] because the regime-conditional diagnostics and stress diagnostics remain centered on that trio.

Allocation Algorithm

The weight construction proceeds in two stages at each rebalancing decision date *t*. Stage 1 computes raw inverse-volatility weights for each asset *i* ($w_{i,raw}$) as:

$$w_{i,raw} = \left(\frac{\sigma_{target}}{\hat{\sigma}_{i,t}}\right) \times \left(\frac{1}{num\_assets}\right), \tag{38}$$

where σ_{target} = 10% annualized is the portfolio volatility target, $\hat{\sigma}_{i,t}$ is the model's forecast of 20-day forward realized volatility for asset *i* at date *t*, and *num_assets = 14* is the number of assets. Before inversion, the forecast row is clipped below at a portfolio-construction floor of *0.02* annualized. The 2% floor is a portfolio safeguard only; it does not alter

[19] executed as matched-window H3 test
[20] 151 features, including ML PCA latent factors
[21] 24 rebalancings, 21-day intervals, 10% annualized vol target, Ledoit–Wolf shrinkage, 10 bps one-way costs
[22] EWMA, XGBOOST, unconstrained MLP, unconstrained LSTM, causal constrained XGBOOST, causal constrained MLP, and causal constrained LSTM
[23] EWMA, XGBOOST, and causal constrained XGBOOST

the prediction artifacts. The floor becomes necessary once the seven-model portfolio layer includes neural network forecasts: unconstrained MLP produces 1,366 below-floor observations in the backtest matrix, unconstrained LSTM produces 317, causal constrained MLP produces 654, and causal constrained LSTM produces 424, while EWMA, XGBOOST, and causal constrained XGBOOST never breach the floor.

Stage 2 estimates ex-ante portfolio volatility via the Ledoit–Wolf shrinkage covariance matrix estimated from the 63 log-return observations immediately preceding the decision date [6]. With num_assets = 14 assets and 63 observations, the ratio $\frac{p}{n} \approx 0.22$ places the Ledoit–Wolf estimator in a range where shrinkage toward the scaled identity matrix has a material effect on the off-diagonal elements, pulling pairwise correlations toward zero relative to the sample covariance and benefiting portfolio diversification. If $\sigma_{portfolio} > \sigma_{target}$, then a portfolio-level scaler is applied to all risky-asset weights with the residual $(1 - \Sigma w_i)$ allocated to a zero-return cash position. The formula of the portfolio-level scaler is defined in Equation (39).

$$portfolio\_scaler = min\left(1, \frac{\sigma_{target}}{\sigma_{portfolio}}\right), \tag{39}$$

where $\sigma_{portfolio}$ is the portfolio volatility**.** The 10% target operates as a ceiling rather than as a leverage-seeking equality constraint because leverage is disallowed. Low-volatility portfolios can remain meaningfully below target when the inverse-volatility risky sleeve already produces ex-ante risk beneath the cap.

Execution and Drift Convention

Rebalancing decisions are made at the close of date *t*; the resulting weight vector becomes active at the open of date *t+1*, eliminating same-day look-ahead bias. Between rebalancing dates, portfolio weights drift with daily simple returns through a mark-to-market mechanism. Each asset's weight *i* on day *d* equals its gross asset value divided by total portfolio value, where total portfolio value is the sum of every asset's gross asset value on that day:

$$w_{\{i,d\}} = \frac{\left[w_{\{i,d-1\}}\, x\, \left(1 + r_{\{i,d\}}\right)\right]}{\left[w_{\{cash,d-1\}} + \sum_{j=1}^{num\_assets} w_{\{j,d-1\}}\, x\, \left(1 + r_{\{j,d\}}\right)\right]}, \tag{40}$$

where $w_{\{i,d\}}$ is the post-drift weight of asset *i* on day *d*; $w_{\{i,d-1\}}$ is the prior day's weight; $r_{\{i,d\}}$ is asset *i*'s daily simple return on day *d*; $w_{\{cash,d-1\}}$ is the prior day's weight of cash; and *num_assets* is the number of assets. Drift causes the risky-weight sum to migrate away from the target allocation over the 21-day holding segment; the monthly rebalancing event restores the target weight vector on the next execution day. Daily simple returns computed as $\exp(\log return) - 1$ are used exclusively for portfolio PnL compounding; log returns are retained for Ledoit–Wolf covariance estimation, consistent with standard practice for daily-frequency portfolio simulation.

Transaction Costs

Transaction costs of 10 basis points one-way[24] are applied on the execution day, which is the first day of each holding period, computed as:

$$Transaction\ cost = transaction\ cost\ one\ way \times \Sigma\left|w_{i,new} - w_{i,pre-trade}\right|, \tag{41}$$

where $w_{i,pre-trade}$ reflects the drift-adjusted weight carried forward from the previous segment and $w_{i,new}$ reflects the target post-rebalance weight for asset *i*. No transaction costs are assessed on non-rebalancing days. The cost model applies identically across all seven model variants, ensuring that cross-model performance differences reflect forecast quality rather than asymmetric cost assumptions. The inception allocation is treated as a portfolio launch event: pre-trade weights are zero, so the inception cost equals $transaction\ cost\ one\ way \times \Sigma w_{i,new}$.

[24] 20 bps round-trip

Backtest Fairness and Governance

All seven model variants share an identical backtest protocol: the same test-partition calendar[25], the same 24 rebalancing events, the same 63-day Ledoit–Wolf covariance lookback, the same 10% volatility target, the same 10 bps one-way transaction cost, and the same 2% forecast floor in the portfolio-construction layer. The only experimental variation is the source of the per-asset volatility forecast $\hat{\sigma}_{i,t}$: EWMA uses the span-63 exponentially weighted moving standard deviation, XGBOOST, unconstrained MLP, and unconstrained LSTM use the unconstrained forecasts, and causal constrained XGBOOST, causal constrained MLP, and causal constrained LSTM use the DAG-constrained forecasts. This shared-engine design ensures that observed portfolio performance differences are attributable to the forecast family rather than to protocol asymmetries [18].

The refactored backtest produces a richer set of outputs than the earlier three-model portfolio chapter. The weight table records the full allocation diagnostic chain at each of the 24 rebalancing events for all seven model variants. The daily returns table captures daily gross return, net return, equity curve, drawdown, and 21-day rolling realized volatility across 499 active trading days per model variant.

Chapter 4: Results

4.1 H1 — Volatility Forecast Accuracy – Comparison between unconstrained and constrained XGBoost

In this section, we evaluate whether applying DAG-based causal constraints to XGBoost improves out-of-sample volatility forecasting accuracy compared with both the unconstrained XGBoost model and the EWMA benchmark. The results are presented in Figure 10 and Tables 9–12.

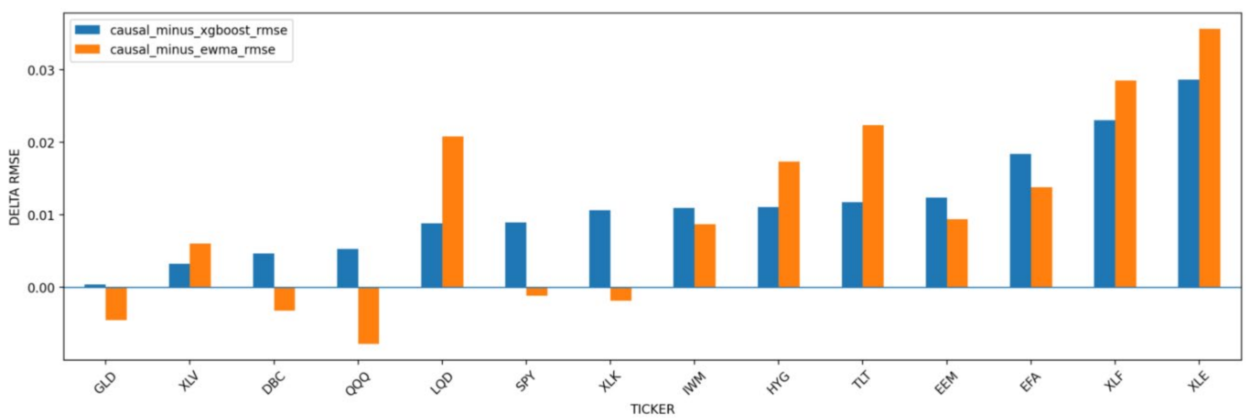

Figure 10. Causal Model RMSE vs. Unconstrained XGBoost and EWMA.

Table 9. Relative Performance of Causal-Constrained XGBoost, XGBoost, and EWMA.

Asset class	Avg RMSE Δ (Causal Constrained XGBoost– Unconstrained XGBoost)	Avg RMSE Δ (Causal Constrained XGBoost – EWMA)	Avg MAE Δ (Causal Constrained XGBoost – Unconstrained XGBoost)	Avg MAE Δ (Causal Constrained XGBoost –EWMA)
COMMODITY	0.00254	-0.00384	0.00151	-0.00286
US EQUITY	0.00839	-0.00008	-0.00108	-0.00667
FIXED INCOME	0.01053	0.02017	0.00314	0.00909
INTERNATIONAL EQUITY	0.01538	0.01159	0.00094	0.00249
SECTOR EQUITY	0.01639	0.01707	0.00066	0.00457

[25] 2023-12-29 through 2025-12-31, 499 active trading days

Table 9 reveals that, across asset classes the DAG constraints increase RMSE relative to unconstrained XGBoost. Against EWMA, the constrained model improves commodities and roughly ties U.S. equities, but underperforms in fixed income, international equity, and sector equity. EWMA is a volatility benchmark, not a momentum benchmark. In addition, it can be analyzed that causal model compared to EWMA is struggling the most at fixed income, shown by the worst absolute degradation of RMSE and MAE.

Table 10. Diebold-Mariano Test Results of Causal-Constrained XGBoost, Unconstrained XGBoost, and EWMA.

Loss Type	Model A	Model B	Mean Loss Diff (A–B)	DM Statistic	P-Value (2-sided)	Winner
Squared Error	Causal XGBoost	XGBoost	0.0019	0.978	0.328	XGBoost
Squared Error	Causal XGBoost	EWMA	0.0015	0.758	0.448	EWMA
Absolute Error	Causal XGBoost	XGBoost	0.0010	0.240	0.811	XGBoost
Absolute Error	Causal XGBoost	EWMA	0.0018	0.352	0.725	EWMA

The volatility forecast accuracies are compared using the Diebold Mariano Test across three models[26]. We chose only these three models because they represent the primary comparisons of this project[27]. The neural network models are included in the portfolio construction for completeness, but they are not the primary focus of our claims. Focusing on three models also helps with the interpretation of the results and directly ties to the hypothesis 1.

Table 10 shows that the DAG-constrained model has higher point-estimate losses, but the Diebold-Mariano HAC tests do not reject equal predictive accuracy, which mean the RMSE penalty is not statistically significant over the available test window. H1 is, therefore, not confirmed. In addition, DM Statistics are all below 1.0, which mean the forecast distributions between the causal model and non-causal model are near-identical. It can also be inferred that performance gap between the causal XGBoost and non-causal XGBoost shrink further under absolute error; the causal model's underperformance is caused by few large errors, not systematic bias. Tables 11 and 12 present the RMSE and MAE, respectively, for predicting volatility in all seven models, which gives a clearer view of overall model performances.

Table 11. RMSE of EWMA, Unconstrained LSTM, MLP, and XGBoost, and Constrained LSTM, MLP, and XGBoost (Bold Value: Best Performing Model).

Model	LSTM	MLP	EWMA	XGBoost	Causal_XGBoost	Causal_MLP	Causal_LSTM
Aggregate Equal Weight	0.095	0.167	0.071	**0.070**	0.081	0.175	0.088
DBC	0.069	0.169	0.048	**0.040**	0.045	0.150	0.059
EEM	0.102	0.163	0.072	**0.069**	0.081	0.187	0.089
EFA	0.090	0.165	0.073	**0.068**	0.087	0.161	0.087
GLD	0.084	0.175	0.068	**0.063**	0.064	0.143	0.086
HYG	0.041	0.139	**0.027**	0.033	0.044	0.154	0.038
IWM	0.112	0.168	0.084	**0.082**	0.093	0.164	0.098
LQD	0.042	0.149	**0.021**	0.033	0.042	0.148	0.048
QQQ	0.134	0.176	0.108	**0.095**	0.100	0.216	0.112
SPY	0.118	0.181	0.099	**0.089**	0.098	0.197	0.108
TLT	0.057	0.150	**0.032**	0.043	0.054	0.148	0.055
XLE	0.131	0.183	**0.102**	0.109	0.138	0.256	0.137
XLF	0.126	0.174	**0.083**	0.088	0.111	0.162	0.100
XLK	0.134	0.188	0.120	**0.108**	0.119	0.196	0.127
XLV	0.090	0.162	**0.053**	0.056	0.059	0.161	0.091

[26] EWMA, unconstrained XGBoost and causal-constrained XGBoost
[27] EWMA vs. unconstrained ML vs. causal constrained ML

Table 11 shows that EWMA and unconstrained XGBoost lead on aggregate RMSE; the MLP family is the weakest across the board. Causal Constraints XGBoost adds modest RMSE versus its unconstrained counterpart. These RMSE increases occurred across all tickers, although the increases were relatively modest for GLD, QQQ, and XLV compared to the other assets. It can be hypothesized that the smaller RMSE increases are caused by the characteristics of these three indices, which are driven by relatively stable, well-understood, and economically coherent factors.

Table 12. MAE of EWMA, Unconstrained LSTM, MLP, and XGBoost, and Constrained LSTM, MLP, and XGBoost (Bold Value: Best Performing Model).

Model	LSTM	MLP	EWMA	XGBoost	Causal_XGBoost	Causal_MLP	Causal_ LSTM
Aggregate Equal Weight	0.070	0.131	**0.048**	0.049	0.050	0.122	0.062
DBC	0.050	0.135	0.036	**0.031**	0.036	0.106	0.045
EEM	0.081	0.130	**0.051**	0.053	0.052	0.130	0.065
EFA	0.067	0.128	**0.043**	0.045	0.047	0.115	0.062
GLD	0.065	0.141	0.050	0.046	**0.044**	0.099	0.066
HYG	0.031	0.109	**0.018**	0.021	0.025	0.110	0.024
IWM	0.083	0.133	0.063	0.059	**0.054**	0.120	0.065
LQD	0.028	0.117	**0.015**	0.020	0.022	0.105	0.031
QQQ	0.094	0.138	0.074	0.068	**0.066**	0.149	0.080
SPY	0.084	0.139	0.064	**0.058**	0.061	0.144	0.068
TLT	0.045	0.121	**0.026**	0.036	0.039	0.108	0.042
XLE	0.092	0.137	**0.061**	0.070	0.074	0.167	0.092
XLF	0.098	0.137	**0.056**	0.064	0.064	0.109	0.071
XLK	0.092	0.140	0.083	0.075	**0.074**	0.136	0.092
XLV	0.064	0.128	**0.036**	0.042	0.041	0.111	0.065

Table 12 reveals that the MAE pattern mirrors the RMSE pattern at lower magnitude. Causal Constrained XGBoost aggregate MAE (0.050) is essentially indistinguishable from the unconstrained XGBoost MAE (0.049), even though the RMSE gap is wider. However, it also can be inferred that EWMA performs marginally better against unconstrained XGBoost. Causal constraints consistently improve LSTM on both RMSE and MAE, while MLP shows a split MAE improvement on average, but RMSE deteriorates in MLP, which indicates that causal structure reduce MLP's typical errors but amplifies its tail mistakes, whereas LSTM with its recurrent architecture is able to reduce the tail impacts.

4.2 H2 — Portfolio Risk Control

We evaluate portfolio risk control across all seven model variants by examining risk-adjusted returns, drawdown behavior, turnover, and transaction costs. The analysis treats the DAG's effect on the decision-tree and neural-network families separately, and it closes with a focused look at the April 2025 tariff shock, the steepest drawdown episode in the backtest window.

4.2.1 Model Performance Summary

Table 13 presents the full portfolio performance summary across all seven models, covering return, risk, drawdown, and cost metrics.

Table 13. Performance summary of EWMA, Unconstrained LSTM, MLP, and XGBoost, and Causal-Constrained LSTM, MLP, and XGBoost (bold red = worst-performing model; bold green = best-performing model).

Model	EWMA	XGBoost	MLP	LSTM	Causal_XGBoost	Causal_MLP	Causal_LSTM
Annualized Return	8.67%	8.10%	**6.07%**	**11.40%**	8.73%	6.36%	9.50%
Annualized Volatility	6.93%	6.11%	7.18%	**8.55%**	**6.05%**	6.87%	8.02%
Sharpe Ratio	0.583	0.567	**0.227**	**0.777**	0.667	0.275	0.611
Max Drawdown	-7.38%	-6.03%	-8.18%	**-9.10%**	**-5.83%**	-8.58%	-8.69%
Calmar Ratio	1.174	1.343	0.742	1.253	**1.498**	**0.741**	1.094
Final Equity	1.179	1.167	**1.124**	**1.238**	1.180	1.130	1.197
Total Transaction Cost	**0.003**	0.004	0.013	**0.014**	0.004	0.011	0.012
Mean Cash Weight	0.189	0.246	0.278	0.132	0.242	**0.291**	**0.083**
Mean Risky Weight Sum	0.811	0.754	0.722	0.868	0.758	**0.709**	**0.917**
Avg Realized Vol 21d	6.44%	5.83%	6.41%	**7.72%**	**5.82%**	6.05%	7.49%

Within the EWMA vs. unconstrained-XGBOOST vs. causal-constrained-XGBOOST comparison, causal-constrained XGBOOST has the best Sharpe ratio, the shallowest maximum drawdown, and the highest Calmar ratio. Across all seven models, BASELINE_LSTM holds the highest full-period Sharpe, while causal-constrained XGBOOST still holds the shallowest full-period drawdown and highest Calmar. Therefore, H2 is partially confirmed.

4.2.2 Equity Curves of Unconstrained and Constrained Models

An equity curve tracks the cumulative value of a portfolio over time, and we compare the equity curves of the unconstrained and causal-constrained model portfolios as shown in Figure 11 to assess how DAG constraints shape returns trajectories across the backtest period.

Figure 11 shows that the seven portfolios move closely together through most of 2024 and clearly diverge during (and after) the April-May 2025 stress episode. The unconstrained LSTM model finishes highest, while the causal constrained LSTM and the causal constrained XGBoost model track more closely in the causal panel. Visually, EWMA and unconstrained XGBoost track so closely together that their equity curves are not distinguishable. Throughout the entire test period, the causal constrained MLP equity curve lags behind the other models and never leads at any point in time.

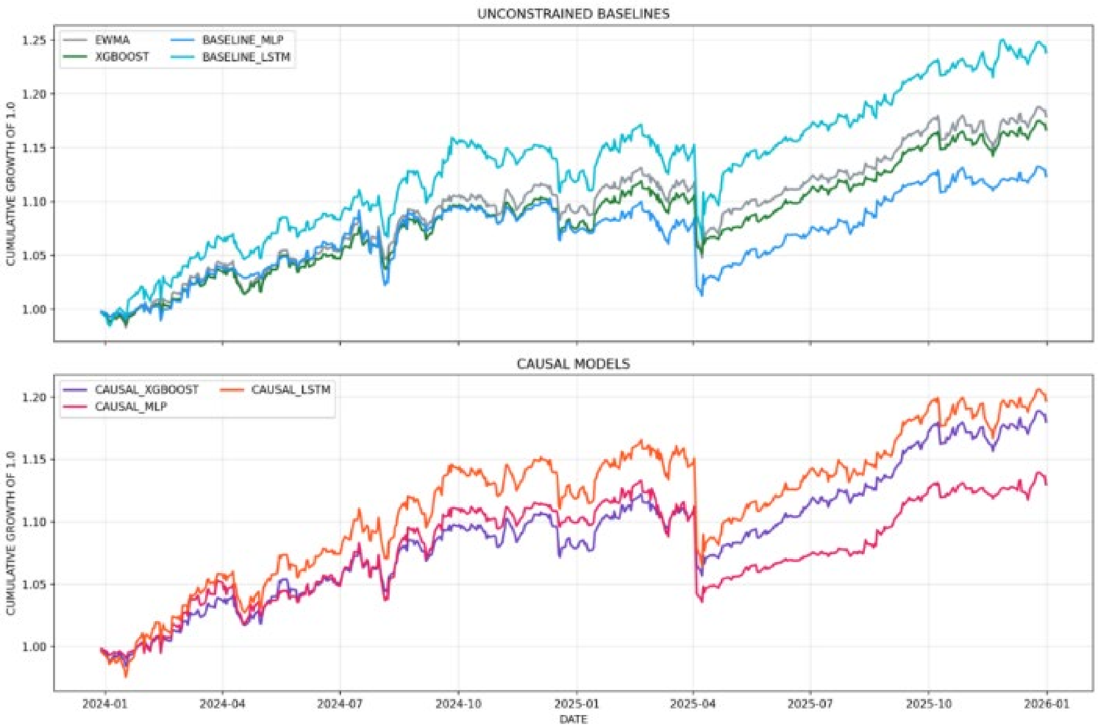

Figure 11. Equity Curves of EWMA, Unconstrained LSTM, MLP, and XGBoost, and Causal-Constrained LSTM, MLP, and XGBoost.

The unconstrained LSTM model scores the highest Sharpe Ratio, final equity, and annual return, but with deepest drawdown and highest transaction cost. The DAG constraints applied to the LSTM model reduce: the annual return to 9.5% (from 11.4% in the unconstrained model), the realized volatility to 7.49% (from 7.72% in the unconstrained model), the Sharpe ratio to 0.611 (from 0.777 in the unconstrained model), the Calmar Ratio to 1.094 (from 1.253 in the unconstrained model), and the final equity to 1.197 (from 1.238 in the unconstrained model).

The causal constraints applied to the MLP model improve the annual return, the Sharpe ratio, and the final equity and reduce the annualized volatility compared to the unconstrained MLP model. The causal constraints happened to increase the drawdown compared to the unconstrained MLP model and happened to decrease the annualized volatility by smaller margin, unlike the reduction observed in the LSTM model. The simple EWMA still beats causal constrained LSTM and causal constrained MLP in terms of Calmar Ratio.

Applying DAG constraints to the XGBoost model improves four primary risk and risk-adjusted metrics relative to the unconstrained XGBoost baseline:

- The Sharpe ratio improves from 0.567 to 0.667.
- Annualized volatility decreases from 6.11% to 6.05%.
- The Calmar ratio improves from 1.343 to 1.498.
- The 21-day realized volatility decreases from 5.83% to 5.82%.

The DAG constraints remove features that cause forecast overfitting to short-term signals. As slightly less accurate but steadier estimate leads to smoother, lower turnover portfolio adjustment, which translates better risk adjusted return after cost. The DAG's effect on the neural-network families is more mixed.

4.2.3 Drawdown of Unconstrained and Constrained Models

Drawdown measures the peak-to-trough decline in portfolio value at any point in time, capturing the worst-case loss an investor would have experienced. Figure 12 plots the drawdown trajectories of all seven models over the full backtest window.

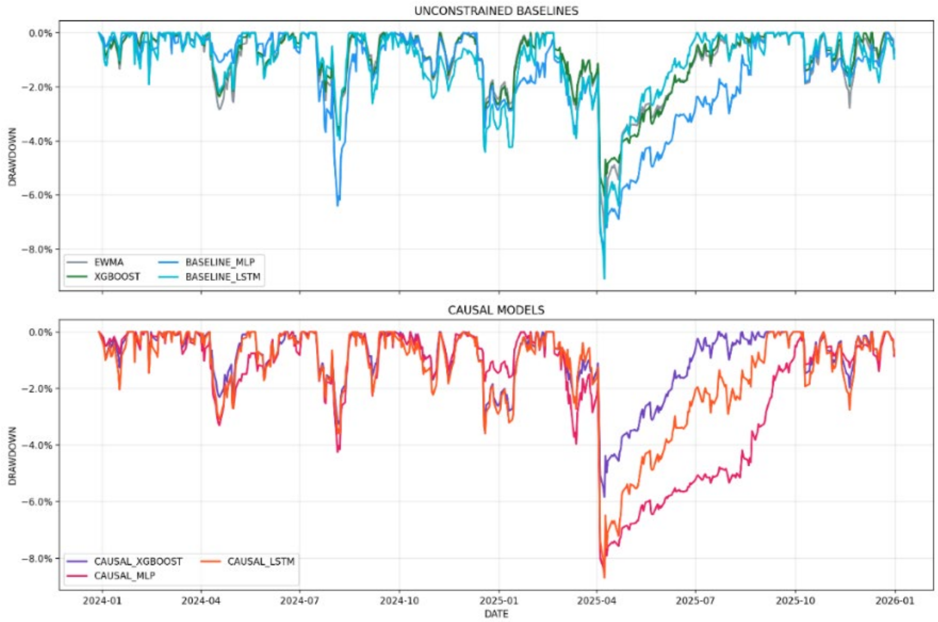

Figure 12. Drawdown of EWMA, Unconstrained LSTM, MLP, and XGBoost, and Causal-Constrained LSTM, MLP, and XGBoost.

Figure 12 reveals that during the stress period, the drawdown of causal constrained LSTM and causal constrained MLP curves are more pronounced compared to the causal constrained XGBoost; this is supported by their annual volatilities (8.02% vs. 6.87% vs. 6.05%). The causal constrained neural network models produced more volatile equity growth, where the causal constrained LSTM model grows the fastest but experiences steep drawdown; likely driven by its high allocation to risky assets (mean risky weight sum: 0.917) and relatively low cash holdings (mean cash weight: 0.083). Meanwhile, the causal constrained MLP model moves closely with the causal constrained XGBoost model initially, then performs poorly after the stress episode and remains in deeper drawdown territory well into Q2 2025. At the same time, the causal constrained XGBoost model moves closer to the causal constrained LSTM model after the stress period. Overall, the causal constrained XGBoost is able to grow its equity more sustainably.

4.2.4. Asset Weights of the Causal Constrained XGBoost Model Portfolio

Asset weights are adjustable proportion of assets inside a portfolio. To understand how the causal constrained XGBoost model achieves its risk control, Figure 13 shows how portfolio weights are allocated across the top six assets and cash over the backtest window.

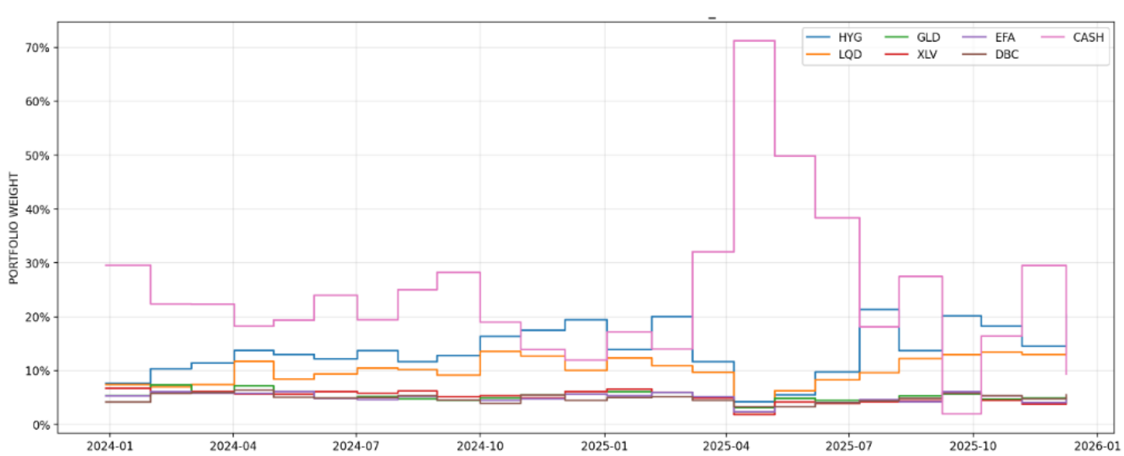

Figure 13. Top 6 Asset Weights + Cash of the Causal Constrained XGBoost Model Portfolio.

Figure 13 reveals that during the stress period, the causal constrained XGBoost model asset allocation shifts towards cash[28]. This explains XGBoost model's drawdown resilience. HYG and LQD (fixed income) dominate causal constrained XGBoost non-cash allocation, which exhibits this model's tendency for conservative positioning.

4.2.5. Average Turnover, Ex-Ante Volatility, and Transaction Cost of Unconstrained and Constrained Models' Portfolio

Average turnover is the mean change in portfolio weights between rebalances. Ex-ante volatility is the model's forecast of portfolio risk. This section examines portfolio turnover, ex-ante volatility, and transaction costs across all seven models, as these metrics capture how frequently each model rebalances, the forward-looking risk it targets, and the trading costs it incurs. Figure 14 and Table 14 summarize these metrics.

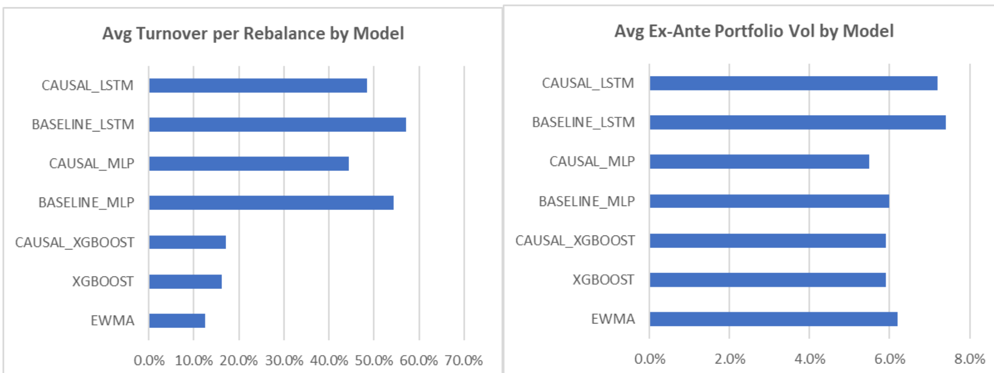

Figure 14. Average Turnover per Rebalance by Model and Average Ex-Ante Portfolio Volatility by Model.

Figure 14 reveals that the EWMA model, the unconstrained XGBoost model, and the causal constrained XGBoost model achieve low turnover (~13-17%). Low turnover often means that these models will lean toward a buy and hold strategy. The MLP and LSTM models exhibit 4-5 times higher turnover (~45-57%), which infers high active trading. The ex-ante plot reveals that the causal constrained LSTM model and the unconstrained LSTM model have volatilities above 6%, the most volatile profiles of all the models. The DAG constraints applied to the MLP model are able to suppress ex-ante volatility (6%→5.5%).

Table 14. Turnover and Transaction Cost Summary of EWMA, Unconstrained LSTM, MLP, and XGBoost, and Causal-Constrained LSTM, MLP, and XGBoost.

Model	Mean Turnover	Median Turnover	Max Turnover	Total Turnover	Total Transaction Cost	Mean Transaction Cost
EWMA	0.125	0.089	0.768	3.004	0.003	0.01%
XGBoost	0.162	0.122	0.636	3.885	0.004	0.02%
MLP	0.544	0.557	1.127	13.058	0.013	0.05%
LSTM	0.572	0.582	1.000	13.735	0.014	0.06%
Causal_XGBoost	0.172	0.132	0.704	4.130	0.004	0.02%
Causal_MLP	0.445	0.414	0.788	10.677	0.011	0.04%
Causal_LSTM	0.485	0.444	1.000	11.633	0.012	0.05%

Table 14 reveals that the unconstrained LSTM model has the highest mean turnover and total turnover, while the unconstrained MLP model has the highest maximum turnover. The Neural network variants trade more often than the decision tree models and EWMA. The DAG constraints reduce turnover in both neural network families, though the causal constrained XGBoost model's turnover is slightly higher than unconstrained the XGBoost model.

[28] shown by cash graph spikes to ~70% on April 2nd 2025

4.2.6 Stress-Window Analysis - April 2025 Tariff Shock

We zoom in on the April–July 2025 tariff shock episode to isolate model behavior during the sharpest market stress in the backtest window. Figure 15 and Table 15 compare loss trajectories and key risk metrics for EWMA, unconstrained XGBoost, and causal constrained XGBoost over this period.

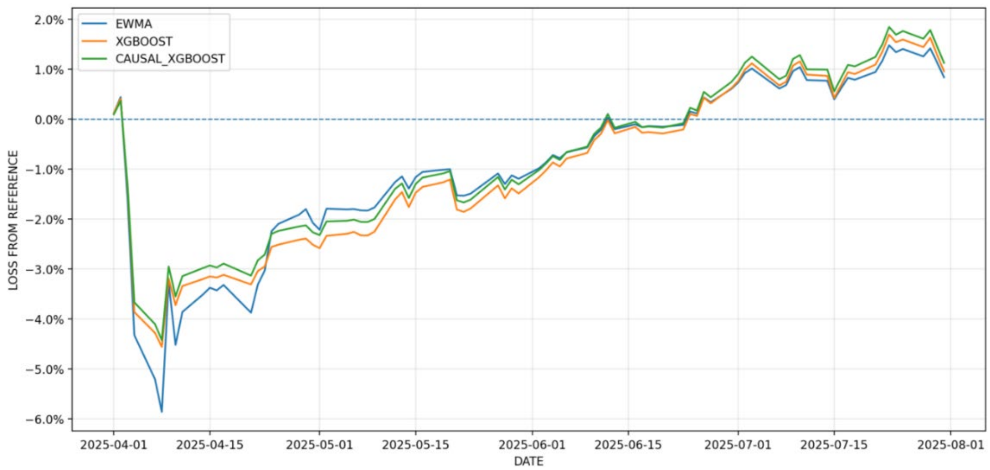

Figure 15. Stress-Window Loss from the 2025-03-31 Reference for the EWMA, the Unconstrained XGBoost Model, and the Causal-Constrained XGBoost Model over the Drawdown Episode.

Table 15. Portfolio Performance over the Stress Window.

Model	EWMA	XGBoost	Causal_XGBoost
Reference Equity	1.113	1.102	1.106
Trough Equity	1.048	1.051	1.057
Stress Loss from 31-Mar-2025 Reference (% from Peak to Trough within Window)	-5.86%	-4.56%	-4.43%
Mean Realized Vol 21d	0.073	0.061	0.060
Max Realized Vol 21d	0.177	0.133	0.130
RMSE	0.0708	0.0698	0.0811
MAE	0.0483	0.0491	0.0501
Sharpe Ratio	0.5830	0.5665	0.6671
Full-Period Max Drawdown	-0.0738	-0.0603	-0.0583
Calmar Ratio	1.1743	1.3432	1.4980

The stress window runs from 1 April 2025 to 31 July 2025, with the steepest drop on 8 April 2025. The portfolio risks are compared across three models. We only chose these three models because they represent the primary comparisons of this project[29]. The neural network models are included in the portfolio construction for completeness, but they are not the primary focus of our claims.

Table 15 reveals that the causal constrained XGBoost model has the lowest drawdown, lowest mean realized volatility, and lowest maximum realized volatility, among the three tree and statistical models. From the 31 March 2025 reference, the drawdown of each of the model are the EWMA = -5.86%, the XGBoost model = -4.56%, and the causal constrained XGBoost model = -4.43%. The constrained XGBoost decision tree also has the lowest mean and maximum realized volatility in the stress window.

[29] EWMA vs. unconstrained ML and unconstrained ML vs. causal constrained ML

The causal constrained XGBoost model produces a 16.2% higher aggregate RMSE than the unconstrained XGBoost model, yet it delivers the shallowest drawdown, highest Sharpe ratio, and highest Calmar ratio. The forecast accuracy and the portfolio risk control do not move together perfectly in this experiment. The causal constrained XGBoost model pays an RMSE penalty but produces the best Sharpe Ratio, shallowest drawdown, and highest Calmar ratio among the EWMA, the XGBoost model, and the causal constrained XGBoost model.

4.3 H3 — NLP Sentiment Contribution

H3 examines whether adding NLP-derived sentiment features to XGBoost improves volatility forecast accuracy. Figure 16 and Table 16 compare RMSE and MAE between the XGBoost model with and without sentiment features across all 14 tickers.

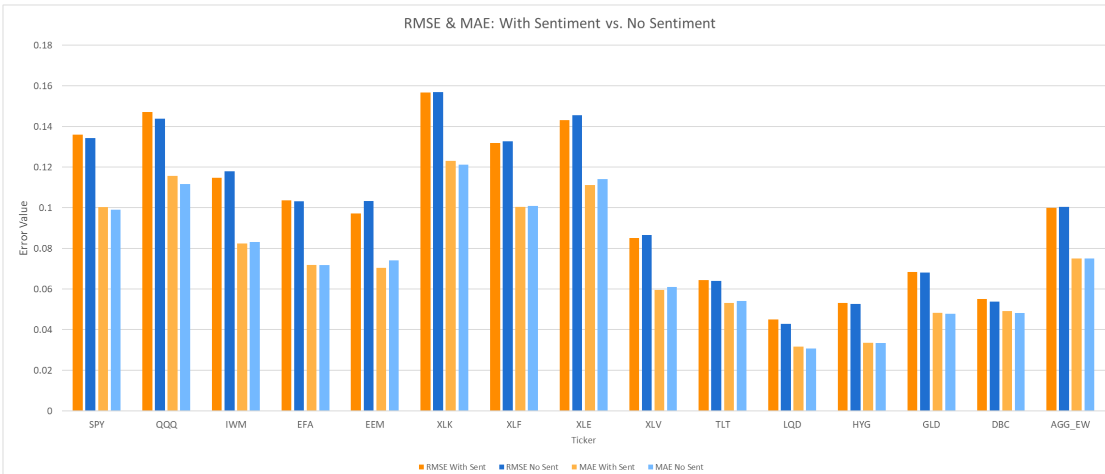

Figure 16. RMSE and MAE Bar Plot of XGBoost without Sentiment and XGBoost with Sentiment.

Table 16. RMSE and MAE of the XGBoost Model with and without Sentiment Data.

Ticker	RMSE of XGBoost with Sentiment	MAE of XGBoost with Sentiment	RMSE of XGBoost without Sentiment	MAE of XGBoost without Sentiment
SPY	0.136	0.100	0.134	0.099
QQQ	0.147	0.116	0.144	0.112
IWM	0.115	0.082	0.118	0.083
EFA	0.104	0.072	0.103	0.072
EEM	0.097	0.071	0.103	0.074
XLK	0.157	0.123	0.157	0.121
XLF	0.132	0.100	0.133	0.101
XLE	0.143	0.111	0.146	0.114
XLV	0.085	0.060	0.087	0.061
TLT	0.064	0.053	0.064	0.054
LQD	0.045	0.032	0.043	0.031
HYG	0.053	0.034	0.053	0.033
GLD	0.069	0.048	0.068	0.048
DBC	0.055	0.049	0.054	0.048
Aggregate Equal Weight	0.100163	0.075	0.100487	0.075

A matched-window evaluation for the XGBoost model with sentiment feature and the XGBoost model without sentiment feature was conducted on an identical post-2020 window (1,496 rows, 300 test observations per ticker). From the graph above, it can be analyzed that per-ticker differences between the XGBoost model with sentiment feature and the XGBoost model without sentiment feature are marginal in both directions.

Table 16 reveals that the XGBoost model with sentiment feature improves aggregate RMSE by -0.000324 (with sentiment 0.10016 versus without sentiment 0.10049), a small 0.32% relative gain. Because no Diebold-Mariano HAC test was run for H3, this result is directionally supported, but not statistically confirmed.

4.4 Secondary Interpretability Diagnostics – Shannon Entropy

Shannon entropy is the evenness of a model's reliance across its features. This section uses Shannon entropy to examine how the DAG constraints change the way the XGBoost model uses features. In this context, higher entropy means the model relies more evenly on many features, while lower entropy means the model depends more heavily on only a few features. Tables 17 and 18, along with Figures 17 and 18, show how the feature importance distribution changes before and after the causal constraints are applied.

Table 17. Top Feature Gain of the Unconstrained XGBoost Model.

Feature	Gain
VOL_XLK_ewma_vol_span21	0.550756037
MOM_XLF_cum_ret_10d	0.408763349
VOL_QQQ_ewma_vol_span21	0.401744634
VOL_SPY_ewma_vol_span21	0.357433856
MACRO_vixcls	0.246976987
REGIME_vix_high	0.223900437
VAL_TLT_hml_beta_63d	0.196913347
MOM_SPY_cum_ret_21d	0.194208711
MOM_SPY_cum_ret_10d	0.179288775

Table 17 reveals that volatility features dominate the unconstrained model's global top-gain ranking, but a momentum feature from XLF lands at rank 2 and a value feature from TLT at rank 7. These cross-asset shortcuts are what the DAG forbids at the Risk node in the constrained version. Table 18 shows the comparison between the unconstrained XGBoost model and the causal constrained XGBoost model in terms of feature importance and entropy.

Table 18. Feature Importance and Entropy Comparison Between Unconstrained and Causal-Constrained Models.

	GLD	SPY	TLT
Entropy Baseline Unconstrained	4.446	4.038	4.039
Entropy Causal Constrained	3.826	3.517	3.602
Normalized Entropy Baseline Unconstrained	0.886	0.805	0.806
Normalized Entropy Causal Constrained	0.889	0.817	0.837
Non Zero Feature Count Baseline Unconstrained	151	151	151
Non Zero Feature Count Causal Constrained	74	74	74
Top 1 Gain Share Baseline Unconstrained	0.063	0.092	0.069
Top 1 Gain Share Causal Constrained	0.093	0.206	0.133
Top 5 Gain Share Baseline Unconstrained	0.204	0.329	0.299
Top 5 Gain Share Causal Constrained	0.296	0.396	0.404
Entropy Delta Causal Minus Unconstrained	-0.620	-0.521	-0.437
Normalized Entropy Delta Causal Minus Unconstrained	0.003	0.012	0.031

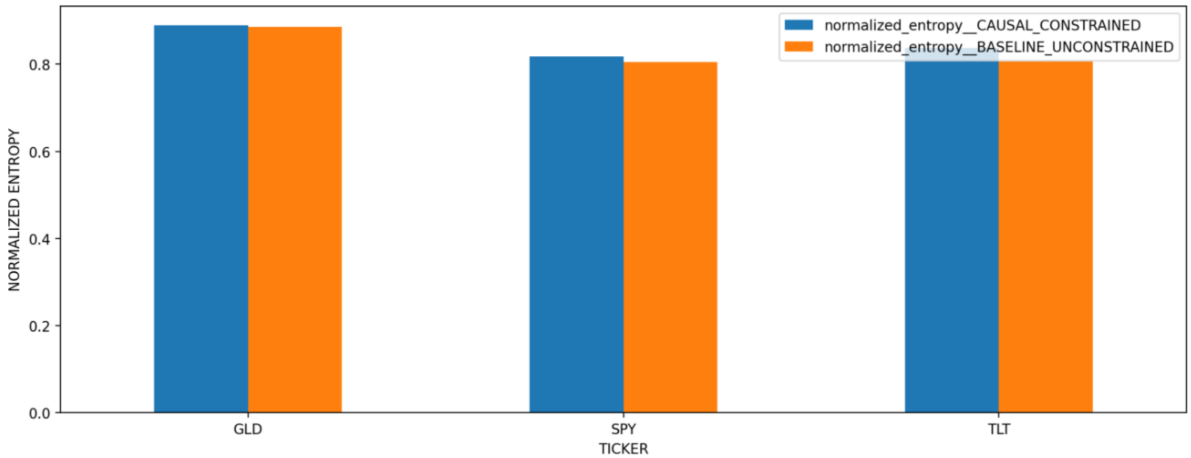

Figure 17. Normalized Feature Importance Entropy Comparison of the Causal-Constrained XGBoost Model vs the Unconstrained XGBoost Model.

Table 18 and Figure 17 show that normalized feature-importance entropy increases under the DAG for GLD, SPY, and TLT, while raw feature-importance entropy decreases. This means the causal-constrained XGBoost model spreads importance more evenly across its allowed 74 features compared with how the unconstrained XGBoost model distributes importance across its 151 features. At the same time, the drop in raw entropy is expected because the constrained model simply has fewer features available (151 → 74), which naturally lowers the maximum possible entropy. Figure 18 shows the changes of prefixes allocation before and after causal constraints applied.

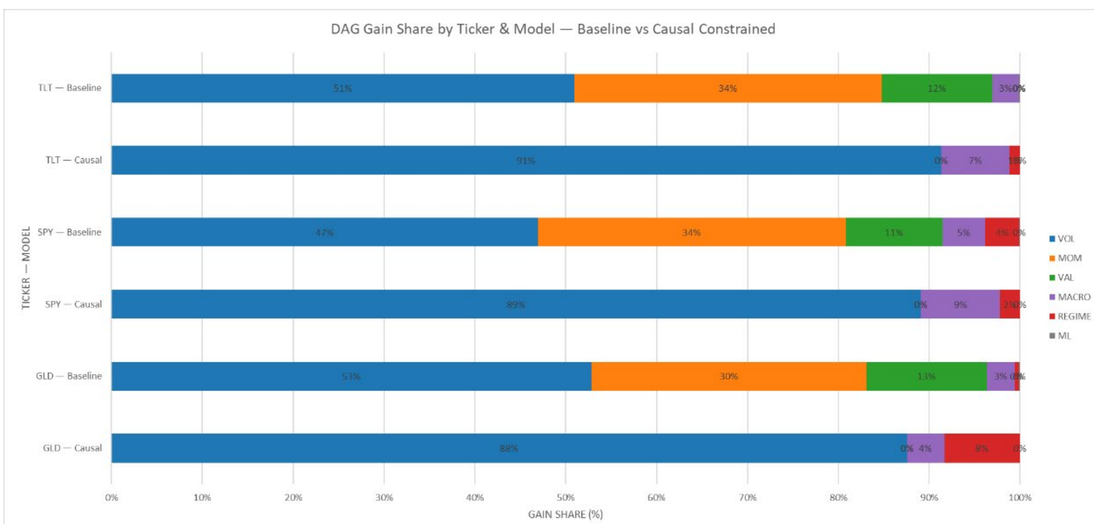

Figure 18. DAG Gain Share Comparison Between the Causal-Constrained XGBoost model and the Unconstrained XGBoost Model by Ticker.

Figure 18 reveals that the VOL (volatility) prefix dominance amplifies after causal constraints applied to the XGBoost model (baseline ~50%, causal ~88-93%). The figure also shows that the regime prefix allocation of TLT jumps significantly to 18%, while SPY/GLD ~8%-14%. It can be inferred that the unconstrained baseline was partially utilizing MOM and MACRO as shortcut proxies[30]. The DAG constraints strip these shortcuts and are able to reveal that VOL (volatility) + Regime prefixes are structurally sound combination. TLT's higher REGIME share also makes sense as bond volatility is more sensitive to macro regime shifts compared to equity volatility.

[30] correlated but not causally valid

4.5 Factor-Exposure Entropy

Factor-exposure entropy measures how evenly a portfolio's risk is distributed across different factor types, including volatility, momentum, value, macro, and regime factors, at each rebalance period. Higher entropy indicates that the portfolio is more balanced across these factors, suggesting more diversified and stable risk exposure. Figures 19 and 20 and Table 19 show how these metric changes over time for EWMA, unconstrained XGBoost, and causal-constrained XGBoost.

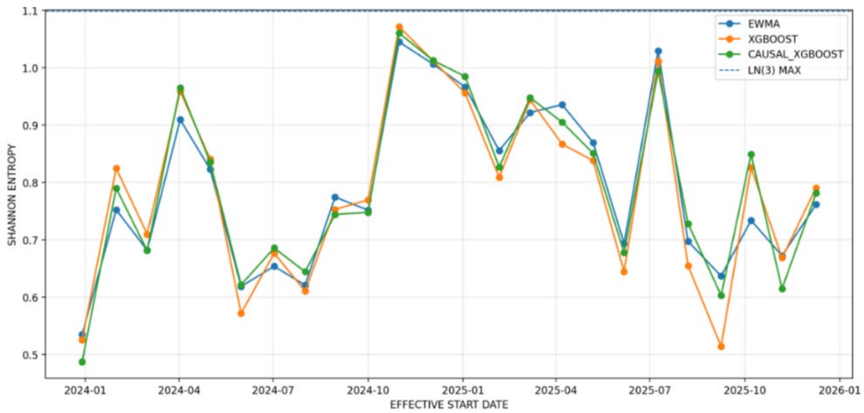

Figure 19. Factor-exposure entropy by rebalance date for the EWMA, the Unconstrained XGBOOST model, and the Causal Constrained XGBOOST model.

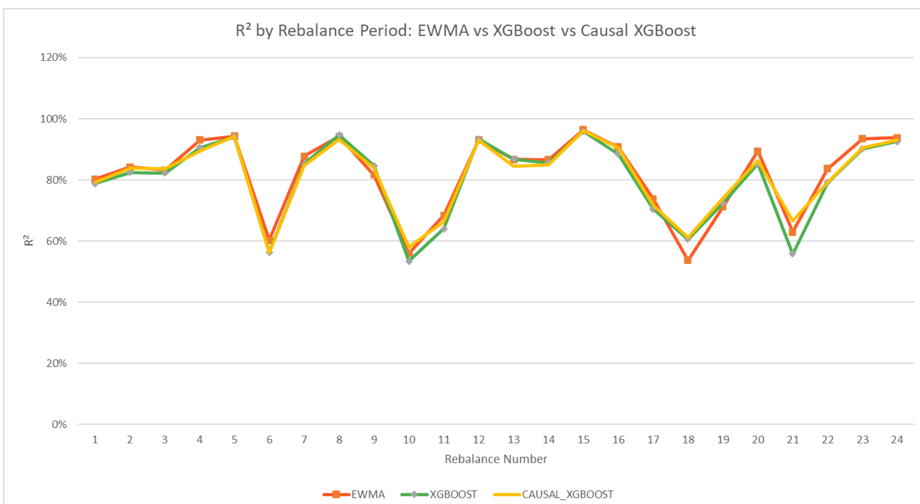

Figure 20. R-squared by rebalance date for the EWMA, the Unconstrained XGBOOST model, and the Causal Constrained XGBOOST model.

Table 19. Average Raw Factor Entropy, Normalized Factor Entropy and R-Squared of the EWMA, the Unconstrained XGBoost model, and the Causal Constrained XGBoost model.

Metric	EWMA	XGBoost	Causal Constrained XGBoost
Avg Factor Exposure Entropy	0.7894	0.7853	0.7933
Avg Factor Exposure Entropy Normalized	0.7186	0.7148	0.7221
Avg R-squared	0.8158	0.8008	0.8096

Table 19 reveals that the causal constrained XGBoost model able to achieves the highest average factor-exposure entropy across all 24-rebalancing segment (0.7933 vs 0.7894 (EWMA) and 0.7853 (unconstrained XGBOOST)). Both tiers point in the same direction, and the DAG prefix gating produces a more uniformly distributed information-loading structure. The causal constrained XGBoost also has the highest mean normalized factor-exposure entropy (0.7221 vs 0.7186 (EWMA) and 0.7148 (unconstrained XGBOOST)). We treat this as directional interpretability evidence, not a material performance gap.

4.6 Regime & Stress Analysis

This section examines how model performance varies across market regimes, splitting the backtest into high-VIX and low-VIX periods to assess whether the DAG constraints provide more consistent behavior during periods of elevated market stress.

Regime Forecasting

Figure 21 and Table 20 compare forecast errors, Sharpe ratios, and maximum drawdown across high-VIX and low-VIX regimes for EWMA, unconstrained XGBoost, and causal constrained XGBoost.

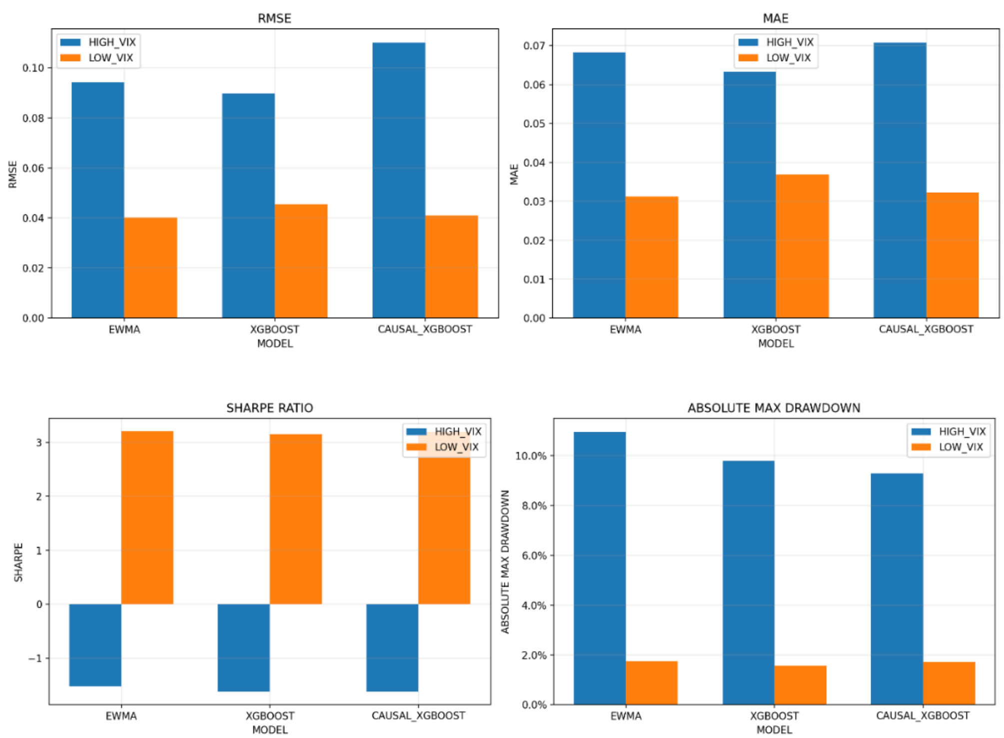

Figure 21. Regime-Conditional Forecast Error and Portfolio Metrics for High VIX versus Low VIX Regimes Across the EWMA, the XGBoost Model, and the Causal Constrained XGBoost Model.

Table 20. Regime-Conditional Forecast Error.

Model	Regime Label	RMSE	MAE
Causal_XGBoost	High -VIX	0.110	0.071
EWMA	High -VIX	0.094	0.068
XGBoost	High-VIX	0.090	0.063
Causal_XGBoost	Low -VIX	0.041	0.032
EWMA	Low -VIX	0.040	0.031
XGBoost	Low-VIX	0.045	0.037

Figure 21 and Table 20 reveal that the forecast errors are higher in the high VIX regime than the low VIX regime for all three models. The unconstrained XGBoost model performs best in the high VIX regime, EWMA performs best in the low VIX regime forecast accuracy, and the causal constrained XGBoost model contributes more through drawdown control than forecast-error dominance. All three models have positive Sharpe ratio in the low VIX regime and negative Sharpe ratio in the high VIX regime. Causal-constrained XGBoost has the shallowest maximum drawdown in the HIGH_VIX regime at −9.29%, versus unconstrained XGBoost at −9.80% and EWMA at −10.95%.

4.7. Risk Targeting

Risk targeting evaluates how closely each model's realized portfolio volatility tracks the 10% annualized volatility target built into the allocation engine. Figure 22 and Table 21 show the realized versus target volatility profiles across all seven models over the full backtest window.

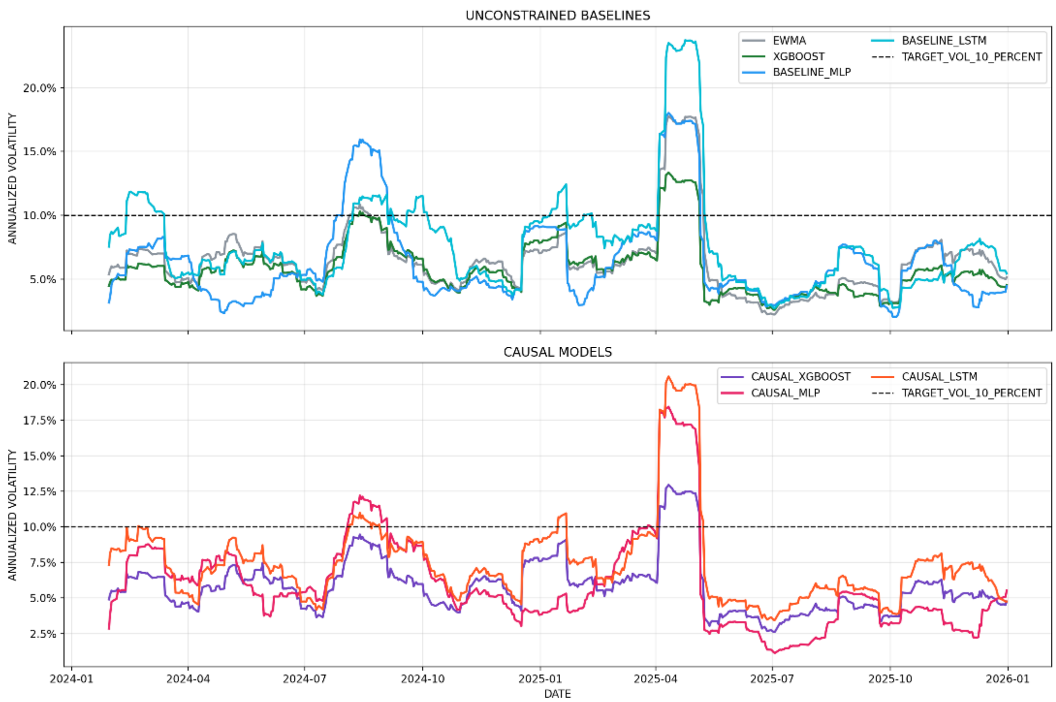

Figure 22. Realized Versus Target Volatility of EWMA, Unconstrained MLP, LSTM, and XGBoost, and Causal Constrained MLP, LSTM, and XGBoost.

Table 21. Volatility Profiles of EWMA, Unconstrained MLP, LSTM, and XGBoost, and Causal Constrained MLP, LSTM, and XGBoost.

Model	EWMA	XGBoost	MLP	LSTM	Causal_ XGBoost	Causal_ MLP	Causal_ LSTM
Target Vol	0.1	0.1	0.1	0.1	0.1	0.1	0.1
Mean Realized Vol 21d	0.064	0.058	0.064	0.077	0.058	0.060	0.075
Median Realized Vol 21d	0.062	0.056	0.052	0.069	0.057	0.053	0.070
Min Realized Vol 21d	0.022	0.026	0.021	0.027	0.026	0.011	0.035
Max Realized Vol 21d	0.177	0.133	0.180	0.237	0.130	0.184	0.206
Mean Abs Deviation from Target	0.042	0.044	0.047	0.037	0.044	0.047	0.034
RMSE VS Target Vol	0.046	0.047	0.051	0.046	0.046	0.053	0.040
Pct Days within 8 to 12 Pct Band	0.077	0.079	0.127	0.317	0.058	0.182	0.271
Mean Ex-Ante Portfolio Vol	0.062	0.059	0.060	0.074	0.059	0.055	0.073

Table 21 shows that all seven portfolios realized volatility in the 5.8% to 7.7% range, remaining well below the 10% target across all models, indicating that the allocation engine behaves more as a capped-risk system than a leverage-seeking volatility-targeting system. Within this lower-volatility regime, the causal constrained models generally maintain more stable portfolio risk outcomes, particularly through lower maximum realized volatility spikes and reduced drawdown behavior. The causal constrained LSTM exhibits the lowest deviation from the target overall, while within the tree-based family, the causal constrained XGBoost model records the smallest realized volatility spike.

4.8. Ablation Summary

A three stage ablation study systematically removes or adds one component at a time to isolate the contribution of each design choice. Stage 1 is the full-window unconstrained baseline. Stage 2 isolates sentiment only in the matched post-2020 forecast window. Stage 3 applies the DAG constraint. Stage 2 is not portfolio-comparable with Stage 1 and Stage 3 because it is forecast-only, uses the matched post-2020 sentiment window, and was not run through the full portfolio backtest. Therefore, portfolio metrics are NA. Table 22 summarizes the results across all three stages.

Table 22. Ablation Summary (Bold = Best Performance).

Stage	Model	RMSE	MAE	Sharpe Ratio	Maximum Drawdown	Calmar Ratio	Normalized Feature Entropy	Factor Entropy
Stage 1 Baseline	XGBoost	**0.0698**	**0.0491**	0.567	-0.0603	1.343	0.832	0.785
Stage 2 H3 Matched Window Sentiment	XGBoost with sentiment	0.1002	0.0751	NA	NA	NA	NA	NA
Stage 3 DAG Constrained	Causal constrained XGBoost	0.0811	0.0501	**0.667**	**-0.0583**	**1.498**	**0.848**	**0.793**

Table 22 shows that applying DAG constraints to the model increases the forecast error. However, they improve Sharpe ratio, maximum drawdown, Calmar ratio, normalized feature entropy, and factor entropy, which indicates that there is a trade-off between forecast accuracy and portfolio performance when applying DAG constraints to the model.

Chapter 5: Conclusion

A proper research process has been executed in this project with 6 reproducible python notebooks. Machine Learning models are powerful but can exploit spurious correlations in historical data. These models are able to discover patterns that look predictive historically but are actually noise. When the market regime changes, these patterns break down and the model fail to deliver good result. The DAG is a governance and information-flow constraint, not a claim about the true data-generating process. This project evaluates whether a manually specified Directed Acyclic Graph (DAG), encoded with economic theory as a hard constraint permitting only features that are treated as economically plausible, improves the performance of machine learning and decision tree models. Improvement is assessed by comparing portfolio performance and forecast accuracy before and after the application of causal constraints. Momentum forecasts near-term returns rather than risk and is therefore excluded from the causal feature set. Value exposure explains returns, not risk. Three hypotheses and one interpretability diagnostics are defined to guide the experiment across 14 ETFs and 500 days out of sample test window. By enforcing this economic theory through prefix-based feature gating, the pipeline produces the following results:

Volatility Forecast Accuracy (H1)

Based on the changes in forecast accuracy, it can be analyzed that by applying causal constraints to the XGBoost model increases forecast error under RMSE and MAE. However, it also should be noted that based on the Diebold-Mariano test shows that this difference is not statistically significant, which means that the two models' results are essentially the near-identical. The performance gap is driven primarily by a small number of large errors rather than consistent bias, as evidenced by the narrow MAE difference. This marginal increase in errors can be traced to the difference in training features between the causal constrained and unconstrained models. The unconstrained XGBoost model occasionally cheats by using non-economically grounded features and producing large calls. In contrast, the causal constrained XGBoost model refuses to cheat and pays a small RMSE penalty.

Against the EWMA model, the causal constrained XGBoost model performs better on commodities and roughly ties on US equities, but falls behind on fixed income, international equity, and sector equity. This performance gap can be traced to the differing characteristics of volatility across asset classes. Fixed income volatility (TLT, LQD, HYG) moves slowly and stays at roughly the same level for extended periods; this means the volatility curves of these three assets are very smooth and persistent. International equities (EFA, EEM) have similar persistent volatility characteristics, but their volatility is partly driven by FX movements and geopolitical noise. Sector equities have idiosyncratic volatility drivers specific to their industries. These characteristics explain why EWMA is able to perform better because it only needs to track recent volatility using exponential decay, while the machine learning model tends to learn too much. Commodity volatility (GLD, DBC) can spike sharply and then collapse episodically; this causes EWMA to struggle and perform worse because it adapts to changes slowly through the decay parameter, which makes it lag badly during sharp transitions. For US equities, VIX is used as one of the features, so the tie between the two models makes sense.

Within the neural network families, the DAG constraints are able to help the LSTM model reduce forecast errors, while for the MLP model they reduce the typical errors but fail to handle the tails. The performance gap likely stems from the fundamental architectural difference between MLP and LSTM, the former being non-recurrent and the latter recurrent. H1 is, therefore, not confirmed.

Portfolio Risk Control (H2)

Based on the portfolio performance, it can be analyzed that the causal constrained XGBoost model produces the strongest risk-adjusted results within the decision tree family, as shown by the increase in Sharpe ratio, reduction in maximum drawdown, and increase in Calmar ratio compared to the unconstrained XGBoost model. As volatility forecasts rose during the April–May 2025 stress episode, the inverse-volatility weights compressed and the portfolio level volatility scaler activated, shifting asset allocation into cash. The causal constrained XGBoost model forecast stream therefore produced the shallowest drawdown trajectory. These results show that the DAG constraints applied to the model make the forecasts more stable, which in turn produces more stable weights, lower turnover, and a portfolio that is able to respond to genuine macro risk signals rather than statistical noise.

For neural networks, it can be analyzed that the effect of applying DAG constraints is more mixed. Applying DAG constraints to the LSTM model reduces annual return, Sharpe ratio, and Calmar ratio, while applying DAG constraints to the MLP model increases annual return and Sharpe ratio but worsens drawdown marginally. These performances are driven by the architectural differences between the LSTM model and the MLP model. The LSTM architecture is built with gates and a more complex structure, which makes it more adaptive and able to self-regulate over time; therefore, the DAG constraints can be said to get in its way. In contrast, the MLP architecture is simpler, which enables the DAG constraints to provide some improvement. However, because of its simpler architecture, the DAG constraints are still unable to help the MLP model handle tail behavior effectively. The DAG constraints able to work best on the XGBoost model because it is vulnerable to spurious correlation which DAG constraints designed to solve. Based on these mixed results, H2 is partially confirmed.

The results reveal a notable divergence: causal constraints fail to improve forecast accuracy, leaving H1 unconfirmed, yet succeed in improving portfolio performance, partially confirming H2. We analyze that this condition is caused by the DAG constraints removing signals, making the model slightly less accurate but producing more stable volatility estimates.

NLP Sentiment Contribution (H3)

Based on the results, it can be inferred that adding NLP features to the XGBoost model is not able to bring any significant improvement to the forecast error. Per-ticker differences go in both directions, and no statistical test confirms the results. These mixed results are driven by the fact that news sentiment is only able to capture positive or negative market mood, not market direction, which explains the marginal improvement. H3, therefore, is directionally supported but not statistically confirmed.

Interpretability

Based on the results, it can be analyzed that the reduction in features caused by the DAG constraints (151 → 74) increases the normalized feature entropy, which means the causal constrained XGBoost model spreads its reliance more evenly across its permitted features. It can also be analyzed that the importance of volatility features increases in the constrained model compared to the unconstrained model. DAG constraints prevent the XGBoost model in exploiting momentum and value signals at volatility forecasting stage. The causal constrained XGBoost model is also able to achieve the highest factor-exposure entropy among all three models, which suggests a more balanced and interpretable portfolio loading structure. DAG constraints therefore directionally improve the interpretability of the model.

Regime Analysis

From the results, it can be analyzed that all three models (EWMA, unconstrained XGBoost, and causal constrained XGBoost) produce higher forecast errors during stress episodes. The XGBoost model is able to produce the best forecast accuracy in the high-VIX regime, while EWMA produces the best forecast accuracy in the low-VIX regime. The causal constraints improve the XGBoost model’s drawdown behavior, enabling the causal-constrained XGBoost model to record the shallowest maximum drawdown in the high-VIX regime.

Overall, causal constraints improve model robustness and model interpretability. Causality did not improve raw forecast accuracy, but the accuracy cost is statistically indistinguishable from noise. The following table is the executive summary of this project:

H1 (Forecast Accuracy) → NOT CONFIRMED The DAG-constrained model does not lower RMSE and does not lower MAE, relative to unconstrained XGBoost. However, the Diebold-Mariano test shows that this penalty is not statistically significant at any conventional level
H2 (Portfolio Performance) → PARTIALLY CONFIRMED The DAG-constrained model produces better risk-adjusted returns, as shown by: • Best Calmar Ratio (XGBoost) • Shallowest Drawdown (XGBoost) Better Stress Recovery
H3 (Sentiment Contribution) → DIRECTIONALLY SUPPORTED Adding NLP sentiment to the unconstrained XGBoost reduces RMSE by a small margin
Secondary Entropy Diagnostic (Shannon Entropy) → SUPPORTED Normalized Feature importance entropy is higher under DAG constraints, and factor exposure entropy is also higher across rebalancing windows.

References

[1] Hsieh, M. (2016). Asymmetric volatility and dynamic asset allocation. *Accounting and Finance Research, 5*(2). https://doi.org/10.5430/afr.v5n2p126

[2] Poon, S.-H., & Granger, C. W. J. (2003). Forecasting volatility in financial markets: A review. *Journal of Economic Literature, 41*(2), 478–539.

[3] Markowitz, H. (1952). Portfolio selection. *The Journal of Finance, 7*(1), 77–91. https://doi.org/10.2307/2975974

[4] Ng, S. L., Chin, W. C., Chong, L. L., & Ng, K. W. (2024). Forecasting the realized volatility of Islamic equities using multivariate HAR-type models. *International Journal of Banking and Finance, 20*(1), 39–67. https://doi.org/10.32890/ijbf2025.20.1.3

[5] Yaro, Y. (2024). Volatility forecasting with component GARCH enhanced machine learning models. [Master's thesis, HEC Montréal].

[6] Ledoit, O., & Wolf, M. (2004). A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis, 88*(2), 365–411. https://doi.org/10.1016/S0047-259X(03)00096-4

[7] Timm, B. (2021). Utilizing machine learning to address noise in covariance and correlation matrices: An application and modification of enhanced portfolio optimisation. [Master's thesis, Copenhagen Business School].

[8] Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics, 31*(3), 307–327. https://doi.org/10.1016/0304-4076(86)90063-1

[9] Christensen, K., Siggaard, M., & Veliyev, B. (2022). A machine learning approach to volatility forecasting. *Journal of Financial Econometrics, 21*(5), 1680–1727. https://doi.org/10.1093/jjfinec/nbac020

[10] Engle, R. F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica, 50*(4), 987–1007. https://doi.org/10.2307/1912773

[11] Teller, A., Pigorsch, U., & Pigorsch, C. (2022). Short- to long-term realized volatility forecasting using extreme gradient boosting. *SSRN Electronic Journal.* https://doi.org/10.2139/ssrn.4267541

[12] Bombari, S., & Mondelli, M. (2025). Spurious correlations in high dimensional regression: The roles of regularization, simplicity bias and over-parameterization [Poster]. *Proceedings of the 42nd International Conference on Machine Learning (ICML 2025).* https://icml.cc/virtual/2025/poster/46042

[13] Pearl, J. (2009). Causality: Models, reasoning, and inference. (2nd ed.). Cambridge University Press.

[14] Spirtes, P., Glymour, C., & Scheines, R. (2000). Causation, prediction, and search. (2nd ed.). MIT Press.

[15] Kaddour, J., Lynch, A., Liu, Q., Kusner, M. J., & Silva, R. (2022). Causal machine learning: A survey and open problems (Version 2). *arXiv.* https://arxiv.org/abs/2206.15475

[16] D'Acunto, G., Bajardi, P., Bonchi, F., & De Francisci Morales, G. (2021). The evolving causal structure of equity risk factors. *Proceedings of the 2nd ACM International Conference on AI in Finance (ICAIF '21)* (pp. 1–8). ACM. https://doi.org/10.1145/3490354.3494370

[17] Michel, A., Arun, A., Sarmah, B., & Pasquali, S. (2025). FinCARE: Financial causal analysis with reasoning and evidence [Preprint]. *arXiv.* https://doi.org/10.48550/arXiv.2510.20221

[18] Bailey, D. H., Borwein, J. M., López de Prado, M., & Zhu, Q. J. (2014). Pseudo-mathematics and financial charlatanism: The effects of backtest overfitting on out-of-sample performance. *Notices of the American Mathematical Society, 61*(5), 458–471. https://doi.org/10.1090/noti1105

[19] Gu, S., Kelly, B., & Xiu, D. (2020). Empirical asset pricing via machine learning. *Review of Financial Studies, 33*(5), 2223–2273.

[20] DeMiguel, V., Garlappi, L., & Uppal, R. (2009). Optimal versus naive diversification: How inefficient is the 1/N portfolio strategy?. *The Review of Financial Studies, 22*(5), 1915–1953.

[21] Moreira, A., & Muir, T. (2017). Volatility-managed portfolios. *The Journal of Finance, 72*(4), 1611–1644.

[22] Fama, E. F., & French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics, 33*(1), 3–56. https://doi.org/10.1016/0304-405X(93)90023-5

[23] Cao, L., & Ren, J. (2022). Machine learning shows that the COVID-19 pandemic is impacting U.S. public companies unequally by changing risk structures. *PLoS ONE, 17*(6), e0269582. https://doi.org/10.1371/journal.pone.0269582

[24] Chen, Z. (2024). Robustness investigation of ML models for stock price prediction during market volatility: A case study of the Japanese stock market amidst the COVID-19 pandemic. *Highlights in Business Economics and Management, 40*, 573–580. https://doi.org/10.54097/3cbhg472

[25] Masini, R. P., Medeiros, M. C., & Mendes, E. F. (2021). Machine learning advances for time series forecasting (Version 3). *arXiv.* https://doi.org/10.48550/arXiv.2012.12802

[26] Loughran, T., & McDonald, B. (2011). When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks. *The Journal of Finance, 66*(1), 35–65. https://doi.org/10.1111/j.1540-6261.2010.01625.x

[27] Bodilsen, S. T., & Lunde, A. (2025). Exploiting news analytics for volatility forecasting. *Journal of Applied Econometrics, 40*(1), 18–36. https://doi.org/10.1002/jae.3095

[28] Bask, M., Forsberg, L., & Östling, A. (2024). Media sentiment and stock returns. *The Quarterly Review of Economics and Finance, 94*, 303–311. https://doi.org/10.1016/j.qref.2024.02.008

[29] Tetlock, P. C. (2007). Giving content to investor sentiment: The role of media in the stock market. *The Journal of Finance, 62*(3), 1139–1168. https://doi.org/10.1111/j.1540-6261.2007.01232.x

[30] Heston, S. L., & Sinha, N. R. (2017). News vs. sentiment: Predicting stock returns from news stories. *Financial Analysts Journal, 73*(3), 67–83. https://doi.org/10.2469/faj.v73.n3.3

[31] García, D. (2013). Sentiment during recessions. *The Journal of Finance, 68*(3), 1267–1300. https://doi.org/10.1111/jofi.12027

[32] Sharpe, W. F. (1966). Mutual fund performance. *The Journal of Business, 39*(1), 119–138.

[33] Andersen, T. G., & Bollerslev, T. (1998). Answering the skeptics: Yes, standard volatility models do provide accurate forecasts. *International Economic Review, 39*(4), 885–905. https://doi.org/10.2307/2527343

[34] Araneda, A. A. (2021). The optimal decay parameter in the EWMA model. *arXiv.* https://arxiv.org/abs/2105.14382

[35] Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics, 7*(2), 174–196.

[36] Misra, A., Si, X., Wen, M., Zhou, T., & Han, X. (2025). Volatility forecasting in financial time-series: A comparative study of GARCH, XGBoost, and LSTM models. *SSRN.* https://doi.org/10.2139/ssrn.5595710

[37] Zhu, H., Bai, L., He, L., & Liu, Z. (2023). Forecasting realized volatility with machine learning: Panel data perspective. *Journal of Empirical Finance, 73*, 251–271.

[38] Zheng, X., Aragam, B., Ravikumar, P., & Xing, E. P. (2018). DAGs with NO TEARS: Continuous optimization for structure learning. *Advances in Neural Information Processing Systems, 31*, 9472–9483. https://arxiv.org/abs/1803.01422

[39] Arun, A., Dimino, F., Agarwal, T. P., Sarmah, B., & Pasquali, S. (2025). FinReflectKG: Agentic construction and evaluation of financial knowledge graphs. *Proceedings of the 6th ACM International Conference on AI in Finance (ICAIF '25).* ACM. https://doi.org/10.1145/3768292.3770363

[40] Fama, E. F., & French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics, 116*(1), 1–22. https://doi.org/10.1016/j.jfineco.2014.10.010

[41] Shannon, C. E. (1948). A mathematical theory of communication. *Bell System Technical Journal, 27*(3), 379–423.

[42] Cao, Z. (2025). NLP forecasting of stock price volatility: A hype-adjusted probability measure and hype index. *SSRN.* https://doi.org/10.2139/ssrn.5276577

[43] Ncume, V., van Zyl, T., & Paskaramoorthy, A. (2022). Volatility forecasting using deep learning and sentiment analysis. *arXiv.* https://doi.org/10.48550/arXiv.2210.12464

[44] Diebold, F. X., & Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics, 13*(3), 253–263.

[45] Cohen, G., Aiche, A., & Eichel, R. (2026). Entropy-augmented forecasting and portfolio construction at the industry-group level: A causal machine-learning approach using gradient-boosted decision trees. *Entropy, 28*(1), 108. https://doi.org/10.3390/e28010108

[46] Lo, A. W. (2004). The adaptive markets hypothesis: Market efficiency from an evolutionary perspective. *Journal of Portfolio Management, 30*(5), 15–29. https://doi.org/10.3905/jpm.2004.442611

[47] Cerqueira, V., Torgo, L., & Mozetič, I. (2020). Evaluating time series forecasting models: An empirical study on performance estimation methods. *Machine Learning, 109*, 1997–2028. https://doi.org/10.1007/s10994-020-05910-7

[48] Wilson, G., Bryan, J., Cranston, K., Kitzes, J., Nederbragt, L., & Teal, T. K. (2017). Good enough practices in scientific computing. *PLOS Computational Biology, 13*(6), e1005510. doi:10.1371/journal.pcbi.1005510.

[49] Sandve, G. K., Nekrutenko, A., Taylor, J., & Hovig, E. (2013). Ten simple rules for reproducible computational research. *PLOS Computational Biology, 9*(10), e1003285. doi:10.1371/journal.pcbi.1003285.

[50] Pineau, J., Vincent-Lamarre, P., Sinha, K., Larivière, V., Beygelzimer, A., d'Alché-Buc, F., Fox, E., & Larochelle, H. (2021). Improving reproducibility in machine learning research. *Journal of Machine Learning Research, 22*(164), 1–20.

[51] Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785–794. https://doi.org/10.1145/2939672.2939785

[52] Hornik, K., Stinchcombe, M., & White, H. (1989). Multilayer feedforward networks are universal approximators. Neural Networks, 2(5), 359–366. https://doi.org/10.1016/0893-6080(89)90020-8

[53] Rumelhart, D. E., Hinton, G. E., & Williams, R. J. (1986). Learning representations by back-propagating errors. Nature, 323, 533–536. https://doi.org/10.1038/323533a0

[54] Diqi, M. (2022). StockTM: Accurate stock price prediction model using LSTM. International Journal of Informatics and Computation, 4(1). https://doi.org/10.35842/ijicom

Appendix

Key values:

Metric	Value	Context
Modeling Window	2,496 rows	2016-01-04 to 2025-12-31
Test Window	500 trading days	2023-12-28 to 2025-12-31 (25 blocks of 20 days)
Num of Rebalances	24	Monthly; every 21 trading days
Features: full / non-sentiment / constrained	152 / 151 / 74	51% feature reduction under DAG constraint; 77 of the 151 non-sentiment features are excluded by the risk-stage gate.
Causal XGBoost RMSE Penalty	16.2%	0.0811 vs. 0.0698 for unconstrained XGBoost
Diebold-Mariano Test Minimum p-value	0.328	Above 0.10 threshold; penalty not significant
Causal XGBoost Sharpe Ratio	0.667	Best of tree-based; 17.7% above XGBoost
Causal XGBoost Max Drawdown	-5.83%	Shallowest of all 7 models
Stress Loss (Apr. 2025)	-4.43%	Causal smallest loss vs. EWMA (-5.86) and XGBoost (-4.56)
Normalized Feature Importance Entropy Delta	+0.015	Causal (0.8476) vs. XGBoost (0.8324); interpretability diagnostic supported
Factor Entropy Exposure Delta	+0.008	Causal (0.7933) vs. XGBoost (0.7853); interpretability diagnostic supported at portfolio level
ETF Universe	14	Spans U.S. equity, int'l equity, sector equity, fixed income, commodity
Sentiment Articles	458,662	January 2020 onward; Alpha Vantage Sentiment API

Glossary of Mathematical Notations

Notation	Meaning
$E[R_p]$	expected portfolio return
w	vector of portfolio weights
μ	vector of expected asset returns
w^T	transpose of the weight vector
${\sigma_p}^2$	portfolio variance
Σ	covariance matrix of asset returns
$w^T\Sigma w$	quadratic form measuring total portfolio risk
Sharpe	annualized Sharpe ratio
$\bar{r}_p$	mean daily portfolio return
$\bar{r}f$	mean daily risk-free rate
σ_p	standard deviation of daily portfolio returns
$\sqrt{252}$	annualization factor for approximately 252 trading days
${\sigma^{20}}_{i,t}$	20-trading-day forward realized volatility for asset *i* at date *t*
$r_{i,t+k}$	daily log return of asset *i* on day *t + k*
$\hat{\sigma}_t^2$	EWMA variance estimate at time *t*
λ	decay parameter
$\hat{\sigma}_{t-1}^2$	prior EWMA variance estimate
${r_{t-1}}^2$	squared return from the previous period
$1 - \lambda$	weight on the most recent shock
$\hat{f}$	fitted forecasting function
$\mathcal{F}$	hypothesis space or model class
N	number of observations
$L(y_i, f(x_i))$	loss function comparing realized target y_i and prediction $f(x_i)$
y_i	realized target for observation *i*
x_i	feature vector for observation *i*
$\hat{f}^{(M)}(x)$	boosted prediction after *M* stages
γ_m	learning coefficient at stage *m*
$h_m(x)$	weak learner or regression tree at stage *m*
x	input feature vector
M	total number of boosting stages
$h^{(l)}$	activations in layer ℓ
$\varphi(\cdot)$	nonlinear activation function
$W^{(l)}$	weight matrix for layer ℓ
$h^{(l-1)}$	output from the previous layer
$b^{(l)}$	bias vector for layer ℓ
f_t	forget gate
$\sigma(\cdot)$	Sigmoid activation function
W_f	Gate weight of forget gate
h_{t-1}	previous hidden state
x_t	input vector at time *t*
b_f	Gate bias for forget gate
$\tilde{c}_t$	Candidate cell state
$tanh(\cdot)$	Hyperbolic tangent activation function
W_c	Gate weight for cell state
b_c	Gate bias for cell state
c_t	Cell state at time *t*
c_{t-1}	Cell state at time *t-1*
i_t	Input gate at time *t*
W_i	Gate weight for input gate
b_i	Gate bias for input gate
o_t	Output gate
W_o	Gate weight for output gate

Notation	Meaning
b_o	Gate bias for output gate
h_t	Hidden state
$\odot$	element-wise multiplication
$P(X_1, X_2, \ldots, X_n)$	joint distribution over all variables
X_i	endogenous variable *i* in the graph
$Pa(X_i)$	parent set of variable X_i
$\prod$	product taken across all variables in the DAG
$f_i(\cdot)$	structural assignment function for variable *i*
U_i	exogenous disturbance term or unobserved noise
$R_{i,t}$	return on asset *i* at time *t*
$R_{f,t}$	risk-free rate at time *t*
α_i	asset-specific intercept or abnormal return
$\beta_{MKT,i}$	market-factor loading for asset *i*
$(Mkt - RF)_t$	market excess return at time *t*
$\beta_{SMB,i}$	size-factor loading for asset *i*
SMB_t	size factor at time *t*
$\beta_{HML,i}$	value-factor loading for asset *i*
HML_t	value factor at time *t*
$\varepsilon_{i,t}$	idiosyncratic residual term
$Cov(r_i, HML)$	covariance between asset return and the HML factor
$Var(HML)$	variance of the HML factor
r_i	return series for asset *i*
S	sample covariance matrix
T	number of observations
r_t	asset returns at time *t*
$\bar{r}$	sample mean return vector
$(r_t - \bar{r})^T$	transpose of the centered return vector
H	Shannon entropy
p_k	normalized share for component *k*
K	number of components, factors, or features
$ln(\cdot)$	natural logarithm
H_{norm}	normalized Shannon entropy
$ln(K)$	maximum possible entropy under equal weighting
$\tilde{S}_t$	raw aggregate sentiment score for calendar day *t*
N_t	number of relevant news articles published on day *t*
j	article index
$s_{j,t}$	sentiment score for article *j* published on day *t*
x_t^{sent}	sentiment feature used by the model on trading day *t*
$\tilde{S}_{t-1}$	aggregate sentiment score from the previous calendar or trading day
$t_{train,end}$	final date included in the training set
$t_{test,start}$	first date of the prediction or test block
$forecast\_horizon$	forecast horizon in trading days
$r_{i,t}$	daily log return of ETF *i* on date *t*
$P_{i,t}$	Price of asset *i* at time *t*
$P_{i,t-1}$	Price of asset *i* at time *t-1*
ddof	Delta Degrees of Freedom
r	Correlation coefficient
$\hat{\sigma}_0^2$	EWMA variance estimate at time *0*
r_0^2	Squared-return at time 0
f_k	independent tree structure
num_trees	total number of trees
$\hat{y}_i$	predicted value
z_i	feature vector for training instance *i*
$\mathcal{H}$	space of regression trees
ℓ	loss function

Notation	Meaning
$\mathcal{L}(\phi)$	Regularized objective function
$\Omega(f^k)$	complexity penalty
J	number of leaves in a tree
$\Omega(f)$	regularization term
γ, η	regularization hyperparameter
κ	leaf weight vector
h^0	input-layer activation vector
h_l	output of hidden layer *l*
W_l	weight matrix for layer *l*
b_l	bias vector for layer *l*
$\hat{y}$	predicted output
W_L	output-layer weight matrix
h_{l-1}	hidden layer *l-1*
b_L	output-layer bias vector
$w_{i,raw}$	raw inverse-volatility weights for each asset *i*
σ_{target}	portfolio volatility target
num_assets	number of assets
$\frac{p}{n}$	ratio of number of assets to the number of observations
$(1 - \Sigma w_i)$	residual cash allocation
$\sigma_{portfolio}$	portfolio volatility
$w_{\{i,d\}}$	post-drift weight of asset *i* on day *d*
$w_{\{i,d-1\}}$	prior day's weight
$r_{\{i,d\}}$	asset *i*'s daily simple return on day *d*
$w_{i,new}$	target post-rebalance weight for asset *i*
$w_{i,pre-trade}$	drift-adjusted weight carried forward from the previous segment
$w_{\{cash,d-1\}}$	prior day's weight of cash
$\tau(\cdot)$	activation function
std (·)	sample standard deviation